

Digesting the Four Colour Theorem: schema-robustness of the ring-14 obstruction, a minimum-interior theorem, and a coloring-mass law

(draft — author list to be completed)

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Abstract

The three machine proofs of the Four Colour Theorem (Appel–Haken 1977; Robertson–Sanders–Seymour–Thomas 1997; and the 2026 near-linear-time proof of [arXiv:2603.24880](#)) are accepted but opaque: what they establish beyond the theorem itself is largely unexamined. We report four verified structural facts obtained by re-deriving these proofs computationally from their published mathematics rather than by running their released code.

First and principally, we show that the failure of the 2026 discharging schema over the ring- ≤ 14 sub-pool is a property of the *rule shapes*, not of the published rule amounts. Fixing the 84 discharging rule shapes of [arXiv:2603.24880](#) and restricting the published 8,200-configuration D-reducible pool to its 5,895 members of ring size ≤ 14 , we certify that *no* nonnegative amount vector $x \in \mathbb{R}_{\geq 0}^{84}$ closes the discharging argument — “certify” in the sense of §1.3: an exact-rational Farkas certificate over a row set that is necessary under two stated hypotheses, not an unconditional proof. The proof is a 12-row irreducible Farkas certificate, verified over exact rationals by a solver-free checker that re-derives every row from its source. The same machinery over the full pool is feasible and the published amounts satisfy every generated row, so the encoding is not vacuous. Two hypotheses are load-bearing and stated as such: nonnegativity of the amounts, and our reconstruction of the degree-5/6 half of the discharge, which the 2026 paper discharges in a single sentence by citation. This bears directly on a question Steinberger [19] states he has “no opinion” about, namely whether a D-only unavoidable set of ring size ≤ 14 exists; we settle it for this pool and these rule shapes, and no further.

Second, we determine $f(r)$, the minimum number of interior vertices of a D-reducible configuration of ring size r , for $r = 8, \dots, 13$: $f = 5, 5, 6, 7, 7, 8$. The bounds are exhaustive (over 13.2 million plantri-generated near-triangulations, 10,319 of which are RSST-legal configurations below threshold, none D-reducible) and tight (explicit catalog witnesses). This is the first quantitative extension of Birkhoff’s 1913 result that no reducible configuration has ring size ≤ 5 .

Third, we report an independent re-verification of the published 4CT configuration corpus by a reducibility checker written from the RSST paper rather than ported from its code — the RSST 633 (249 D-reducible, 384 C-reducible), Steinberger’s 2,822, and 7,697 of the 2026 pool, all with computed $|\Phi(K)|$ matching the published headers exactly — together with an end-to-end reproduction of the 2026 C++ pipeline. Along the way we replaced a legacy C oracle that aborts partway through Steinberger’s set, and identified 430 configurations of the 2026 pool that no stage of the pipeline uses, re-verifying the whole pipeline without them.

Fourth, we develop a “coloring-mass” framework: exact wheel closed form $a = (2^r + 2(-1)^r)/6$, an improved cap $a \leq 11 \cdot 2^{r+k-7}$ tight at every wheel, a proof of $\gamma = 3/2$ at $k = 2$, and an *obstruction theorem* showing that no proof factoring through $a \leq P(S, 4)/24$ can reach the empirically sharp constant $4/3$.

We are explicit about what did not work. A neural component produced a modest boundary-classification result and a feature extraction that failed on one of four folds; it did not discover any theorem here. Three mined sufficient conditions for reducibility were killed by adversarial mutation search, which produced 236 false positives, 215 of them confirmed by an independent implementation; a natural LP formulation was killed by its own negative control; the naive threshold law was killed by the lemma harness. We position the work relative to the Tao/Buzzard “digestion” agenda: the contribution is tooling and verified structural facts that make an opaque machine proof more comprehensible, not an automated discovery.

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1 Introduction

1.1 The digestion problem

The Four Colour Theorem has been proved three times by machine [1, 2, 14, 19, 11] and once in a proof assistant [9]. In every case the proof consists of an *unavoidable set* of configurations, each verified *reducible* by a finite computation, together with a *discharging* argument certifying unavoidability. The certificates are enormous — 633 configurations and 32 discharging rules for RSST [14]; 2,822 configurations and 42 rules for Steinberger [19]; 8,200 configurations and

84 rules for the 2026 near-linear-time proof [11] — and are essentially never inspected for structure. Correctness has been established; comprehension has not.

This paper reports on a programme of *computational digestion*: re-deriving these proofs from their published mathematics with independent implementations, and then mining the resulting data for statements that a human can hold in mind. The framing is Tao’s and Buzzard’s [20, 6]: a machine proof becomes mathematics when it yields transferable structure, and the bottleneck is tooling rather than raw computation. The unit of progress we adopt is a *verified structural fact about the proof objects* — a theorem about the class of reducible configurations, or an impossibility statement about a discharging schema — rather than a shorter proof of the theorem itself.

Throughout, we distinguish sharply between three epistemic grades, and we label every claim in the paper with one:

- **Proven:** a mathematical argument, in some cases with a finite exhaustive case analysis whose completeness argument is given.
- **Certified:** a finite computation whose output is an independently checkable certificate (e.g. a Farkas certificate verified in exact rationals by a program that does not trust the certificate’s own numbers).
- **Corpus-supported:** a statement that survived machine testing over our full labelled corpus of 59,142 configurations. This is *not* a proof, and §9 records what happened the one time we treated it as one.

1.2 The four contributions

(C1) Schema-robustness of the ring-14 obstruction (§3). Steinberger [19] produced the first unavoidable set of *D-reducible* configurations (no contracts), at the cost of ring size up to 16 and 2,822 configurations, and wrote that he had “no opinion” whether an unavoidable set of D-reducible configurations with ring size at most 14 exists. We attack a sharply scoped version of this question. Fix the 84 rule shapes of [11] and the sub-pool P_{14} of the published 8,200-configuration D-reducible pool consisting of the 5,895 configurations of ring size ≤ 14 . Then there is no $x \in \mathbb{R}_{\geq 0}^{84}$ under which the discharging argument closes over P_{14} ; a 12-row irreducible Farkas certificate proves it, verified in exact rational arithmetic without a solver. The contradiction lives entirely at the degree-5/degree-7 interface, and 65 of the 84 rule shapes are irrelevant to it. Because the published amounts satisfy the analogous full-pool system with zero violations, the result distinguishes the two pools rather than being vacuously infeasible.

(C2) The function $f(r)$ (§4). Birkhoff [4] proved that a reducible configuration has ring size at least 6. No quantitative refinement — a lower bound on the *interior* size as a function of the ring size — appears to have been published. We prove

$$f(8), \dots, f(13) = 5, 5, 6, 7, 7, 8,$$

exhaustively and tightly, and note the nuance that the corresponding minimum over *C*-reducible configurations is strictly smaller at rings 8 and 11.

(C3) Independent verification of the corpus (§5). We re-verified the published catalogs with a checker implemented from the RSST paper [16] rather than ported from `reduce.c`, reproduced the 2026 pipeline end to end (four stages, 4h38m wall, asserts live, all counts matching the paper), replaced a legacy C oracle where it aborts, and identified 430 configurations of the 2026 pool that no stage of the pipeline uses.

(C4) The coloring-mass law (§6). Writing $a = |\Phi(K)|$ for the classical RSST count of ring colorings extending to the free completion, we prove the exact wheel value $a = (2^r + 2(-1)^r)/6$ with its equality case, the cap $a \leq 11 \cdot 2^{r+k-7}$ (tight at every wheel), and $\gamma = 3/2$ at $k = 2$. We give an *obstruction theorem*: an explicit 10-vertex configuration shows that no bound factoring through $a \leq P(S, 4)/24$ can prove the empirically sharp cap $a \leq \frac{2^r+2}{6}(4/3)^{k-1}$, the floor for that route being 61/43. Paired with an empirical threshold law, the sharp cap would reproduce the entire $f(r)$ table analytically, to within one interior vertex at every ring.

1.3 What this paper does not claim

We state the limitations up front because several of them are load-bearing.

1. **(C1) does not settle Steinberger’s question.** It is scoped to *these* 84 rule shapes and *this* pool. Other D-reducible configurations of ring ≤ 14 exist outside the published pool, and adding or replacing rule shapes is not covered. Removing the fixed-shapes caveat requires column generation over planar rule patterns (§10).
2. **Unblocked $\neq$ realizable.** Our LP rows are generated for every local structure the pool fails to block, exactly as the released C++ decides which cartwheels to analyse. A structure that is geometrically impossible for a reason outside the pool would yield a spurious row. This is anti-conservative for an INFEASIBLE verdict; it is also precisely the standard the published proof itself uses.
3. **The degree-5/6 rows are a reconstruction.** The released code never enumerates hubs of degree ≤ 6 ; [11] discharges that case in one sentence by citation to [19], who proved it *by machine* at his own amounts. §3.5 gives our reconstruction, its verbatim textual audit against the paper source, and an exhaustive refinement check. Three of the 12 certificate rows depend on it.
4. **The sharp cap is a conjecture**, not a theorem, and the threshold half of the coloring-mass law is corpus-supported only.
5. **No machine-learning discovery is claimed.** §8 reports the neural work honestly, including a replication finding that the same architecture at the same accuracy learned a *different* mechanism than a published result of the same shape.
6. **Schema-incomparability.** Configuration counts across schemas are not directly comparable: RSST’s 633 uses an axle-based discharging schema with C-reducibility and contracts at ring ≤ 14 ; the 2026 schema is wheel/cartwheel-based with D-reducibility only. Any claim about set size must be schema-qualified.

1.4 Related work

On $f(r)$. Birkhoff’s ring-size lower bound [4] is, as far as we can determine, the only published unconditional structural lower bound on the size of a reducible configuration. The literature contains extensive *catalogs* — from which per-ring interior minima can be read off, and which we use as our tightness witnesses — but a catalog minimum is an upper bound on $f(r)$, not a lower bound; the exhaustive negative side of Theorem 4.1 is what is new. We have found no prior claim of a per-ring interior minimum beyond Birkhoff’s ring ≥ 6 , and no prior exhaustive enumeration of the small-interior configuration classes. The nuance that C -reducibility admits smaller interiors (§4.4) is implicit in the RSST catalog but, to our knowledge, has not been stated.

On lower bounds for unavoidable sets. Bounds on the *number* of configurations needed are essentially absent from the literature; the historical sequence ~ 1834 [2] $\rightarrow 633$ [14] $\rightarrow 2,822$ D-only [19] $\rightarrow 8,200$ D-only with a near-linear colouring algorithm [11] records only what has been achieved. Theorem 3.1 is not a bound on set size; it is an impossibility statement about a specific (schema, rule-shape set, pool) triple, and we know of no comparable prior result. The closest antecedent in spirit is Steinberger’s own remark that he has no opinion about the ring-14 question.

On digestion. The programme of making machine proofs comprehensible — rather than merely correct — is articulated by Tao [20] and Buzzard [6], and Gonthier’s Coq formalisation [9] settles the correctness half for the 4CT specifically. Our contribution is on the comprehension side and is deliberately modest in ambition: verified structural facts about the proof objects, plus the tooling that produced them.

On machine learning for mathematical discovery. The template we followed is that of Davies et al. [8, 7]: train a model, probe it, extract a human-statable conjecture. §8 reports our attempt and its partial failure, together with a replication study bearing on the reliability of the “extract a mechanism” step in general. We regard the honest reporting of that attempt as part of the contribution, not as an appendix to it.

1.5 Reproducibility

Every numerical claim in this paper is annotated with the repository artifact that produced it. Appendix A is a claim-to-artifact table giving, for each claim, the file path and what it contains. All third-party inputs (the four arXiv e-prints, the `plantri` generator, the released rule and configuration files, the released C++ checker) are pinned in-repository with recorded SHA-256 digests.

2 Preliminaries

We follow [16, 14] throughout.

2.1 Configurations and free completions

Definition 2.1 (Configuration). A *configuration* K of ring size r with k interior vertices is given by its *free completion* $S = S(K)$: a simple plane near-triangulation of the closed disk on $n = r + k$ vertices such that

1. the outer face is bounded by an induced (chordless) cycle R , the *ring*, of length r , each of whose vertices has degree ≥ 3 ;
2. the *interior* (the k vertices strictly inside) is nonempty and connected, and every interior vertex has degree ≥ 5 ;
3. every bounded face is a triangle.

RSST’s reference parser `ReadConf` in `reduce.c` enforces two further conditions on any file it accepts, and we enforce both:

4. a *labelling convention*: each ring vertex’s neighbour list, written in rotation order, starts at ring-neighbour $i + 1$ and ends at ring-neighbour $i - 1$, with every entry strictly between them interior;
5. the degree identity $\sum_v \deg v = 6(n - 1) - 2r$ (automatic given 1–3);
6. **each interior vertex meets the ring in at most two separate contiguous arcs.**

Condition 6 is genuine structure, not bookkeeping: it is not implied by 1–3, and it removes 33%–77% of the near-triangulations satisfying 1–3 at the sizes we enumerate (§4.2). Condition 4 is a serialization convention only; a cyclic neighbour list denotes the same rotation system whatever entry it is written from. Discovering that distinction cost us a real bug (§7).

2.2 Reducibility

Let $\Phi(K)$ be the set of colourings of $E(R)$ in three colours, taken up to the S_3 action on colours, that extend to a *tri-colouring* of S — an edge 3-colouring in which the three edges of every internal face receive three distinct colours. Set $a = a(K) = |\Phi(K)|$; this is the classical header field of an RSST `.conf` record. K is *D-reducible* if the Kempe-closure operator on subsets of $\Phi(K)$, iterated to its fixed point starting from $\Phi(K)$, reaches $\emptyset$; equivalently, no consistent set survives. K is *C-reducible* if the same holds after contracting a nonempty *contract* X of edges. C-reducibility is strictly weaker: §4.4 exhibits catalog configurations that are C-reducible but not D-reducible with fewer interior vertices than $f(r)$.

Remark 2.2. Everything in §4 is scoped to D-reducibility. The 2026 pool [11] and Steinberger’s set [19] are D-only; the RSST 633 are not.

2.3 Discharging and the three schemas

A discharging argument assigns an initial charge to each vertex of a hypothetical minimum counterexample, moves charge along edges according to a finite set of local *rules*, and derives a contradiction from the fact that total charge is positive while every vertex ends non-positive. RSST use *axles*; Steinberger and [11] use *cartwheels* (a hub with its first and second neighbourhoods, degrees given as intervals). The 2026 paper assigns $T_0(v) := 10(6 - d(v))$, so

$\sum_v T_0(v) = 120$ by Euler ([11] source line 602 and Lemma `lem:120`); a rule R sends $r(R) \in \{1, 2\}$ units from $s(R)$ to $t(R)$; and the final charge is $T(u) = T_0(u) + \sum_{v \sim u} (\phi(v, u) - \phi(u, v))$.

Crucially, *survivors are permitted*: the charge argument need not kill every wheel, because three gluing lemmas mop up the survivors. Encoding “the discharging must kill every wheel” is therefore the *wrong* condition, and §3.10 reports the negative control that demonstrates it: that encoding is infeasible even for the full 8,200-configuration pool, i.e. it would “disprove” the published theorem.

2.4 Computational setting

Our reducibility checker `fourcolor.reduce.check` is implemented from [16]’s mathematics (the `reduce.tex` write-up) rather than ported from `reduce.c`; the 1995 C program is compiled separately as an oracle (`build/reduce_rsst`) purely for differential testing. Exhaustive generation uses `plantri` 5.5 [5], pinned with its SHA-256 digest, invoked as `plantri -Pr -c3 -a n`; with `-P` and no `-m`, `-c3` defaults to `-c3m3`, i.e. no chords and no outer-face vertex of degree 2, exactly matching condition 1. `plantri` is a canonical-construction-path generator producing exactly one representative per isomorphism class, which is what makes “sweep every n below threshold” an exhaustive search rather than a sample.

3 Schema-robustness of the ring- ≤ 14 obstruction

3.1 The question and the exact claim

Steinberger’s unavoidable set [19] is D-only but reaches ring size 16; RSST’s 633 stay at ring ≤ 14 but use contracts. Steinberger states the question and declines to conjecture (`third_party/arxiv-0905.0043/src/4c.tex`, line 137):

“Appel and Haken used only configurations of ring-size at most 14 ... The unavoidable set of Robertson et al. also has only configurations of ring-size 14 or less. Our unavoidable set uses configurations of ring-size up to 16 ... **We were not able to find an unavoidable set using only configurations of ring-size 14 or less, but we have no opinion as to whether such a set exists or not.**”

The 2026 pool [11] is D-only with 8,200 configurations of ring size up to 18, of which 5,895 have ring size ≤ 14 .

A preliminary experiment (§3.2) ran the released 2026 pipeline with the pool restricted to those 5,895: it fails, with 833 of 16,157 per-wheel jobs aborting on an assertion. That is a statement about the *published amounts* x_0 . The result of this section is the much stronger statement that no re-tuning of the amounts can rescue it.

Theorem 3.1 (Certified). *Fix the 84 discharging rule shapes of [11] (`third_party/discharging-rules/R`). Let P_{14} be the sub-pool of the published 8,200-configuration D-reducible pool consisting of the 5,895 configurations of ring size ≤ 14 . Then there is no vector of nonnegative rule amounts $x \in \mathbb{R}_{\geq 0}^{84}$ satisfying the local charge conditions that the 2026 schema imposes over P_{14} : i.e. no $x \geq 0$ under which every cartwheel leaf that P_{14} fails to block satisfies $C(L, x) \leq 0$ (strictly < 0 in the cases of §3.6) and every unblocked degree-5 or degree-6 wheel satisfies $LB(w, x) \leq 0$. In particular the discharging argument of [11] cannot be made to close over P_{14} by re-tuning its amounts. The hypotheses under which*

these rows are necessary conditions — including that an unblocked local structure need not be realizable — are stated in §3.11.

Remark 3.2 (43 figures, 84 rules). [11] draws 43 rules in its Figure `fig:rules`; each is counted twice, once per reflection, except two that are reflection-symmetric, giving $43 \cdot 2 - 2 = 84$ (source lines 635–636, “Figure `fig:rules` shows 84 rules since precisely two, the first and fourth last, are symmetric under the reflection”). This matches the 84 files in `third_party/discharging-rules/R` one for one, and the paper notes the rule set is the same as Steinberger’s 42 except that his 12th and 13th are merged. Our x ranges over all 84 independently, which is weaker than assuming reflection pairs share an amount, hence the safe direction for INFEASIBLE.

The certificate is 12 rows; it is stated in full in Appendix B. Since all 12 rows are non-strict, infeasibility holds over the reals, hence a fortiori over the integers and over the box $\{1, 2\}^{84}$ that [11] allows itself. Section 3.11 lists the hypotheses.

3.2 The empirical precursor: the pipeline on the restricted pool

Before any LP, we simply ran the released pipeline with the pool replaced by P_{14} and the 84 rule shapes and published amounts x_0 left untouched (`tools/s1.build_pool.py`, `results/p3/runs/steinberger-s1/`). The excluded 2,305 configurations are 1,695 at ring 15, 526 at 16, 69 at 17 and 15 at 18.

The schema breaks, and it breaks in a structured way. `combine_rules` yields 681 non-blocked combined rules rather than the baseline 671 — ten combined rules revived, all ten predicted in advance by a gap analysis with 0 unexpected re-blockings. Wheel counts are unchanged at $d = 7, 8$ and grow by 4 and 5 at $d = 9, 10$, all nine revivals again predicted — so the per-wheel job count is $16,148 + 9 = 16,157$ rather than the full pool’s 16,148. Then 833 of 16,157 per-wheel cartwheel jobs abort: 780 on `assert(C == 0)` (530 at $d = 7$, 250 at $d = 8$) and 53 on `assert(d == 7 || d == 8)` (51 at $d = 9$, 2 at $d = 10$). The two failure kinds are perfectly stratified by hub degree, which is itself informative. The run’s verdict is FAIL at the `enum_cartwheels` stage.

This says the published amounts do not work over P_{14} . It does not say no amounts do — and the obvious response, “re-tune x ”, is exactly what Theorem 3.1 rules out. The 833 aborted jobs are also the raw material for the LP: re-run against an NDEBUG build with the assertions compiled out, they yield 77,185 additional fully refined leaves that the assert-guarded run never recorded.

3.3 Encoding: leaf charge is affine in x

The invariant the released C++ actually enforces (`third_party/computer-checks/src/cartwheel.cpp`, `enum_bad_cartwheels`) is, on every fully refined cartwheel L surviving `fix_in_rules` and `fix_out_rules`, the conjunction `assert(C==0)`, `assert(d==7 || d==8)`, `assert(darts_by_deg[7]+[8]+[9] > 0)`. The corresponding necessary condition on x is:

case	requirement	why
$d(L) \in \{9, 10, 11\}$	$C(L, x) < 0$	L must be pruned before the degree assert
$d(L) \in \{7, 8\}$, no spoke of degree ≥ 7	$C(L, x) < 0$	pruned before the spoke assert
$d(L) \in \{7, 8\}$, some spoke ≥ 7	$C(L, x) \leq 0$	$C = 0$ survives to the gluing lemmas

Note that $C = 0$ is *not* imposed: if $C < 0$ the leaf is pruned before the assert is reached, so $C \leq 0$ is exactly the necessary-and-sufficient charge condition at a leaf, weaker than what the C++ observes at x_0 .

At a leaf all d spokes are fixed, so the per-spoke maximum inside `amount_of_possible_charge_send` never runs, and `prune_by_non_associated_rule` forces the spoke combined rules to be determined by the leaf graph. Hence

$$C(L, x) = 10(6 - d) + \sum_k (\text{IN}_k - \text{OUT}_k) x_k, \quad (1)$$

with $\text{IN}_k = \#\{j : \text{always_apply}(\text{dart}_j, R_k)\}$ counting darts pointing into the hub and OUT_k the reverse. The form is affine with integer coefficients: no epigraph variables, no binaries.

Differential validation of (1). All 10,094 published full-pool bad cartwheels in `third_party/computer-checks/wheels/zero` evaluate to *exactly* 0 at x_0 under the Python form, and all 10,094 satisfy $d \in \{7, 8\}$ (9,366 at $d = 7$, 728 at $d = 8$) with a spoke of degree ≥ 7 ; i.e. the Python row semantics independently reproduce all three C++ assertions on the published output, with 0 discrepancies (`tools/lp_discharge.py --validate`).

3.4 Why leaves observed at x_0 constrain every x

The refinement forest is explored with pruning, so the observed leaf set depends on x_0 . The rows are nevertheless valid for all $x \geq 0$:

Lemma 3.3 (Monotonicity). *For every observed leaf L and every $x \geq 0$, the necessary condition $C(L, x) \leq 0$ (resp. < 0 in the strict cases) holds.*

Proof sketch. (a) The branching operations `update_degree_by_rule`, `create_degree_except_tail`, `refine_always` and `refine_never` only narrow degree intervals and never change rotations, so branching is x -independent. It is *not* pool-independent, since `fix_in_rules` branches over the pool-dependent non-blocked combined-rule set — but it moves in the safe direction: a smaller pool blocks fewer combined rules, and we verified that the full-pool combined rules are a content-exact subset of the P_{14} ones (671 $\subset$ 681 files), so the full-pool forest is a sub-forest of the P_{14} forest. (b) `blocked_by_reducible_configuration` is x -independent and monotone in the pool; `prune_by_non_associated_rule` is independent of both. (c) `upper_bound_of_charge` is non-increasing along refinement whenever $x \geq 0$: narrowing degrees can only add `always_apply` matches (out-charge up — this is the step needing $x \geq 0$) and only remove candidates surviving `never_apply`. (d) Hence for any $x \geq 0$, either every ancestor survives and the assert demands $C(L, x) \leq 0$, or some ancestor A has $C(A, x) < 0$ and then $C(L, x) \leq C(A, x) < 0$ by (c). $\square$

An earlier draft asserted flat pool-independence in (a). That was false and was caught by the adversarial pass (§7); the corrected statement is what the argument needs and it holds. Since $P_{14} \subseteq P_{\text{full}}$ (verified: 0 of the 5,895 names absent from the 8,200; both are symlinks to the same files), every full-pool leaf row is also valid for the ring- ≤ 14 problem, which is what lets the certificate mix rows of both origins.

Remark 3.4 (Direction of the relaxation). The row set omits the three gluing lemmas and omits every leaf the x_0 -pruned search never reached. Both omissions make the encoded feasible set *larger* than the true one. Therefore INFEASIBLE transfers to the true problem, while FEASIBLE is only a non-refutation.

3.5 The degree ≤ 6 half — without which the LP is vacuous

The released code enumerates hubs of degree 7–11 only, where the initial charge $10(6 - d)$ is already negative. Taken alone the leaf rows are satisfied by $x = 0$: send nothing and every hub keeps its negative initial charge. We measured exactly that — the “no low-degree” variants in Table 1 are FEASIBLE and vacuously so.

The 2026 paper does not check $d \leq 6$ by hand or otherwise. It asserts the *stronger* $T(v) = 0$ and discharges the obligation in one sentence by citation, inside the proof of its main theorem (source line 1144):

“...we are still missing the case where v has degree at most 6 and positive final charge. However, this cannot happen, for if v has degree 5 or 6, then it has final charge 0 as proved in [19], where exactly the same discharging rules as here are used.”

And [19] does not prove it by hand either: he proves it by machine, with one presentation file per hub degree from 5 to 11, and explicitly declines to write the hand argument (`third_party/arxiv-0905.0043/src/4c.tex`, line 552). So the $d \leq 6$ case rests on a computation at Steinberger’s amounts and against Steinberger’s unavoidable set. When the LP re-tunes x , that citation no longer applies as proved — which is exactly why we impose the necessary condition $T(v) \leq 0$ directly as a row rather than inheriting the conclusion.

For $d \in \{5, 6\}$ we add rows built from a *lower* bound on the hub’s final charge, valid for any $x \geq 0$ and at any level of refinement:

$$\text{LB}(w, x) = 10(6 - d) - \sum_i \text{POSS}_i(x) + \sum_j \text{ALW}_j(x) \leq T(v) \leq 0, \quad (2)$$

where POSS_i sums x_k over base rules *not never apply* out of spoke i (an over-count of sends) and ALW_j sums x_k over base rules that *always apply* into the hub from spoke j (an under-count of receipts). Rows are generated over every spoke-degree necklace in $\{5, \dots, 9\}^d$ up to rotation that the pool does not block.

Textual audit. We audited (2) quote by quote against the paper’s pinned LaTeX source. Initial charge $10(6 - d)$: verbatim, line 602. Total charge 120: Lemma `lem:120`. $T = T_0 - \text{out} + \text{in}$: Definition `dfn:finalcharge`. Requirement at $d \in \{5, 6\}$: the paper asserts $T(v) = 0$, which is *stronger* than our $T(v) \leq 0$, so our rows are conservative. A degree-5/6 vertex with $T = 0$ is a permitted survivor, matching our non-strict rows. Rule $R(1)$ decoded from the paper’s own figure source (`tikz/rule.tex`, lines 967–977) is a single edge with source of degree exactly 5,

target of degree ≥ 5 , amount 2, matching `rule001.rule` byte for byte; so a degree-5 vertex sends 2 along every incident edge unconditionally, and the row $10 - 5x_{\text{rule001}} \leq 0$ is forced. Full audit: `results/steinberger/degree56-verification.md`.

The coarse-cartwheel gap, and its closure. The low-degree rows are built on the *coarse* cartwheel (hub plus spoke-degree necklace, second neighbourhood left at $[5, 9]$), and blocking is tested on that coarse object. “Refining only strengthens the bound” does *not* cover the danger, because refining can also *delete the row*: if a coarse wheel is unblocked but every full refinement is blocked, then no vertex of a minimum counterexample carries that necklace and the row is spurious. That is anti-conservative for INFEASIBLE, so it was checked rather than assumed (`tools/lowdeg_refinement_check.py`).

No low-degree row is spurious. Two findings, the second strictly stronger than needed. (i) Exact enumeration of the certificate’s four degree-5 necklaces over the full refinement space $\{5, \dots, 9\}^m$, $m = 10\text{--}12$, by compiling the blocking predicate into a DNF over the open slots and counting models exactly by DFS; the compiled predicate is differentially checked against the production `wheel_is_blocked` on 2,000 random refinements per necklace (0 disagreements) and exhaustively on all 625 refinements of a 4-slot necklace.

necklace	slots	refinements	blocked	unblocked	surviving
d5-55767 (IIS)	10	9,765,625	5,678,597	4,087,028	41.9%
d5-57577 (IIS)	11	48,828,125	30,711,497	18,116,628	37.1%
d5-57677 (IIS)	12	244,140,625	94,647,089	149,493,536	61.2%
d5-55677 (19-row support)	10	9,765,625	6,350,608	3,415,017	35.0%

Not a knife-edge: between a third and two-thirds of all refinements of each necklace remain unblocked. (ii) The gap closes uniformly for all 86 low-degree rows at once: `PseudoConfiguration::representative_degree` collapses any vertex with degree upper bound > 8 to the single value 9, so the coarse blocking test *is* the blocking test of the refinement pinning every second neighbour to 9. Verified mechanically over the whole sweep: all 580 unblocked $d = 5$ and all 2,466 unblocked $d = 6$ necklaces have an unblocked tail-maximised refinement, 0 exceptions of 3,046.

Sanity anchor. At x_0 , the maximum of LB over all 3,046 unblocked low-degree wheels is exactly 0, never positive; it is attained e.g. at $d = 5$ with all five spokes of degree 9, where the row is literally $10 - 5x_{\text{rule001}} \leq 0$ and x_0 has $x_{\text{rule001}} = 2$. So x_0 sits simultaneously on the boundary of the low-degree rows and of all 3,425 distinct full-pool leaf rows — the signature of a tightly optimised scheme, and a strong consistency check: had we got the $d \leq 6$ side wrong in the loose direction, x_0 would have violated it. Independently, running the released C++ with `--enum_wheels -d 5` and `-d 6` returns exactly 580 and 2,466 unblocked wheels, matching our generation wheel for wheel.

3.6 Code-necessary versus math-necessary

This is the objection that nearly killed the result. `upper_bound_of_charge` counts only `always_apply` rules outward, and `fix_out_rules` stops refining once no rule is `dominantly_apply-but-not-always_apply`. But `dominantly_apply` is strictly stronger than

Table 1: LP variants. Rows are deduplicated by (coefficients, constant, strictness). Reproduce with `tools/lp_discharge.py --all` (≈ 14 min, ≈ 2 GB peak).

variant	rows	strict	x_0 viol.	status
control , full pool, integral, with $d \leq 6$	3,511	0	0	feasible
control, full pool, real, with $d \leq 6$	3,511	0	0	feasible
control, full pool, integral, no $d \leq 6$	3,425	0	0	feasible (vacuous)
control, full pool, real, no $d \leq 6$	3,425	0	0	feasible (vacuous)
main , ring ≤ 14 , integral, with $d \leq 6$	48,509	96	9,647	infeasible ($y^\top b = -30$, 19 rows)
main , ring ≤ 14 , real, with $d \leq 6$	48,509	96	9,551	infeasible ($y^\top b = -30$, 19 rows)
main, ring ≤ 14 , integral, no $d \leq 6$	48,423	96	9,647	feasible (vacuous, $x = 0$)
main, ring ≤ 14 , real, no $d \leq 6$	48,423	96	9,551	feasible (vacuous, $x = 0$)
control, <code>--exact-only</code> (§3.6)	2,620	0	0	feasible
main, <code>--exact-only</code>	42,121	96	8,699	feasible — <i>rows dropped, too lossy</i>
control, <code>--conservative</code>	4,321	0	0	feasible
main, <code>--conservative</code>	52,400	96	8,741	feasible — <i>rows weakened, too lossy</i>

`has_intersection`, so a leaf can retain rules that are neither `always_apply` nor `never_apply` on an out-dart — rules that might fire in some realization but are not counted in `out_charge_sum`. Measured: 11 of 5,292 (rule, out-dart) pairs across the certificate’s leaves are undetermined. For those leaves C strictly *over*-estimates the hub’s true final charge, so $C(L, x) \leq 0$ would be a condition of the released *verification procedure* rather than of the argument itself.

Two blunt diagnostics confirm this is not hypothetical — **both make the ring- ≤ 14 LP feasible** (Table 1, last four rows). Both, however, are needlessly lossy: one throws rows away, the other weakens rows that did not need weakening.

The repair. Pass to the *tail-maximised* leaf: narrow every open tail $[a, 9]$, $a < 9$, to $[9, 9]$. Since 9 means “9 or more” in the cartwheel abstraction, this restricts attention to the sub-family of realizations in which every unpinned second neighbour has large degree. If the tail-maximised leaf (1) has a fully determined OUT side, (2) yields the identical row — same coefficients, same constant, same strictness — and (3) is still unblocked by the pool, then in those realizations OUT is exact while IN can only exceed the branch’s flag sum, so the true final charge is $\geq C(L, x)$, and $T \leq 0$ forces $C(L, x) \leq 0$. Undetermined IN rules are harmless precisely because they push the true charge up.

All 12 certificate rows pass this audit. The nine degree-7 rows satisfy (1)–(3) individually, including the five whose original leaves had undetermined out-pairs; the three degree-5 rows are mathematically necessary by construction, being already in lower-bound form. Machine-readable: `results/steinberger/iis_math_necessity.json`, `all_mathematically_necessary: true`.

3.7 Results

Leaf inputs for the main run: 97,860 fully refined cartwheels yielding 48,423 distinct leaf rows, to which the 86 distinct low-degree rows are added, giving 48,509. Composition of the leaves: 10,094 published full-pool leaves; 10,581 leaves from the 15,324 per-wheel jobs that

completed; 77,185 leaves recovered from the 833 jobs that aborted on an assert, by re-running them against an NDEBUG build with the asserts compiled out (833/833 succeeded, 21 min wall, 8-way parallel). Rows by hub degree: $d=7$: 40,257; $d=8$: 8,070; $d=9$: 93; $d=10$: 3; $d=5$: 26; $d=6$: 60 (the 3,046 low-degree wheels collapse to only 86 distinct rows). The 93 $d=9$ and 3 $d=10$ rows are exactly the strict rows corresponding to the $51 + 2$ `degree_range` failures of §3.2.

Positive control. The full-pool LP is feasible and x_0 satisfies every one of the 3,511 generated rows, with 0 violations. The control was also run without the $d \leq 6$ rows to demonstrate that the trivially feasible regime exists and is not what we report.

Why the headline is the audited certificate. At x_0 , the $\text{ring} \leq 14$ system is violated in 9,647 rows; but x_0 still violates 8,741 rows even in the fully conservative system, with LB up to +3. So at x_0 the $\text{ring} \leq 14$ argument fails not merely as a verification artifact but genuinely, at thousands of degree-7/8 hubs that really do retain positive charge. The `--exact-only` and `--conservative` diagnostics lose the contradiction, which is why the headline is carried by the audited 12-row irreducible subsystem rather than the raw 48,509-row verdict.

3.8 The 12-row irreducible core and what it says

The 19-row Farkas support (Appendix B) is not minimal. Greedy deletion filtering shrinks it to a 12-row *irreducible infeasible subsystem*: nine degree-7 hub rows and three degree-5 rows, irreducible in the strict sense that deleting any one alone restores feasibility (verified for all 12). Its certificate has $y^\top b = -20$ with multipliers 75, 75, 25, 75, 55, 60, 15, 55, 25, 75, 168, 219. All four $\text{ring} \leq 14$ -only rows survive into the core. This is the smallest human-inspectable object carrying the whole result, and it is legible:

- **No degree-8, 9 or 10 row appears.** The contradiction needs none of the 53 `degree_range` failures nor any of the 250 degree-8 failures observed empirically. This sharpens the empirical picture: the irreducible cause sits entirely at hub degree 7, colliding with the degree-5 discharge.
- **Every one of the nine degree-7 hubs has a degree-5 spoke.** The whole contradiction lives in the degree-5/degree-7 interface.
- **Only 19 of the 84 rules carry a nonzero coefficient anywhere in the 12 rows** (rule001, rule002_{1,2}, rule003_{1,2}, rule004_{1,2}, rule008_{1,2}, rule009_{1,2}, rule010_{1,2}, rule011_{1,2}, rule023_{1,2}, rule032_{1,2}). **65 of the 84 shapes are irrelevant to the obstruction** — re-tuning them cannot help.
- In the Farkas combination 78 of the 84 rule columns cancel exactly; only six carry slack. The near-total cancellation is why the contradiction is tight.
- Read as arithmetic: the degree-5 rows say a degree-5 vertex must give away all 10 units of charge, forcing the amounts on rule001–rule004 *up*; the degree-7 rows cap what a degree-7 hub with degree-5 neighbours may receive back, pulling them *down*. With the $\text{ring} \leq 14$ pool no longer blocking four offending degree-7 cartwheels, the two demands cannot both be met.

The infeasibility is genuinely caused by the ring- ≤ 14 -only rows. Delete the four ring- ≤ 14 -only rows from the 19-row support and the remaining 15 — every one of which is also valid for the full pool — are feasible, and x_0 satisfies all 15. The contradiction is not lurking in the full-pool constraints; it is created by exactly the cartwheels the ring- ≤ 14 pool fails to block. Independently, those four rows come from wheels `d7_3358`, `d7_3372`, `d7_3376` — precisely three of the wheels whose job the C++ aborted with `Assertion failed: (C == 0)`. The Python row semantics and the C++ asserts agree wheel by wheel on the certificate’s own support.

3.9 Independent verification of the certificate

`tools/verify_farkas.py` is solver-free and uses `fractions.Fraction` throughout — no floating point anywhere on the checking path. Critically it does not trust the certificate’s stored coefficients: it re-derives every row from its source (`.cartwheel` files re-parsed and re-run through `leaf_row`; low-degree tags rebuilt with `generate_cartwheel` and re-run through `lowdeg_row`) and fails loudly on any mismatch. It then checks (a) $y \geq 0$, (b) $y \neq 0$, (c) $(y^\top A)_k \geq 0$ for all 84 columns, (d) $y^\top b < 0$, prints the contradiction chain $0 \leq (y^\top A)x = y^\top (Ax) \leq y^\top b = -30 < 0$, and independently confirms that x_0 violates 4 of the 19 support rows. Deliberate corruption of a single coefficient is rejected with an exact claimed-versus-re-derived diff.

3.10 The strong encoding, and why it is not used

The alternative encoding — *every stage-1 wheel the pool does not block must have `upper_bound_of_charge` < 0* — is what a naive reading suggests and is what max-of-affine cutting-plane machinery is for. It is *not* a correct encoding of “the argument closes”: survivors are permitted, and the published proof at x_0 leaves $5,439 + 6,790 + 3,285 + 626 + 8 = 16,148$ stage-1 wheels alive on purpose. Imposing it would reject the published proof itself.

We ran it anyway as a negative control, and it fails exactly as predicted: the *full-pool* run is **INFEASIBLE** at iteration 0 with a verified 10-row certificate ($y^\top b = -175$; support: three degree-5, two degree-6 and five degree-7 rows). So the strong condition cannot be met by any nonnegative amounts even with the complete 8,200-configuration pool — it would “disprove” the published theorem. That is the concrete demonstration that the strong encoding is the wrong one. It is reported as a measurement and never as evidence for the headline.

3.11 Scope

1. **These 84 shapes only.** Nothing rules out a different or larger set of rule shapes. Deciding that requires column generation over planar degree-interval patterns, whose pricing subproblem is a search over planar patterns rather than an LP (§10).
2. **This pool only.** P_{14} is exactly the ring- ≤ 14 part of the published pool. Many D-reducible configurations of ring ≤ 14 are not in it.
3. **The $d \leq 6$ rows are a reconstruction** (§3.5); they are load-bearing, since without them the LP is feasible.
4. **Unblocked $\neq$ realizable** (§1.3).

5. $x \geq 0$ is a **hypothesis**, used in Lemma 3.3(c) and in (2). Monotonicity was measured to hold on 2,145 parent/child pairs with $x \geq 0$ and to *fail* on 8 of them when negative amounts are allowed, so the hypothesis is exactly load-bearing, not decorative. In the other direction there is nothing to hedge: the paper fixes $r(R) \in \{1, 2\}$ for all 84 rules, and $\{1, 2\}^{84} \subset \mathbb{Z}_{\geq 0}^{84} \subset \mathbb{R}_{\geq 0}^{84}$, so deciding infeasibility over the nonnegative reals is strictly stronger than the paper’s own amount space requires.
6. **Integrality is not needed:** all certificate rows are non-strict and the real variant is infeasible with the same certificate.

Adversarial pass. The formulation was handed to a reviewer prompted to refute it, defaulting to “the encoding is wrong”, across ten named attack lines. Outcome: one real error found and fixed (the pool-independence claim in Lemma 3.3(a)); one genuine scope correction adopted (the code-necessary versus math-necessary distinction, which became §3.6); the rest survived with evidence. A later pass against the paper’s own LaTeX source found two prose errors (both corrected: the paper does *not* do the degree-5/6 case by hand, and the e-print *was* obtainable) and one genuine open gap (the coarse cartwheel issue), which was closed exactly and exhaustively without changing the certificate. Ablations: the degree-6 rows alone with all leaf rows are feasible; the degree-5 rows alone are already infeasible. Six solver configurations all report infeasible, and a uniform slack of 0.082 on every right-hand side is the minimum needed to restore feasibility.

4 $f(r)$: the minimum interior size of a D-reducible configuration

4.1 Statement

Theorem 4.1 (Proven, exhaustive). *For each ring size $r \in \{8, 9, 10, 11, 12, 13\}$, every D-reducible configuration of ring size r has at least $f(r)$ interior vertices, where*

r	8	9	10	11	12	13
$f(r)$	5	5	6	7	7	8

and each bound is tight: a D-reducible configuration with exactly $f(r)$ interior vertices exists in the published catalogs.

Birkhoff [4] proved that a reducible configuration has ring size ≥ 6 . To our knowledge Theorem 4.1 is the first published lower bound on interior size as a function of ring size.

4.2 The enumeration

For each r and each n from $r + 1$ to $r + f(r) - 1$ we generated *every* near-triangulation of the disk with outer face of size r on n vertices with `plantri -Pr -c3 -a n`, filtered to the RSST configuration class (nonempty connected interior of minimum degree 5; RSST conditions 4–7 including the ring-contact-arc condition 6), and ran the D-reducibility checker on each. Table 2 reports the counts.

Two remarks on the class definition, both of which cost real debugging.

Table 2: The exhaustive sweep. “raw” = near-triangulations from `plantri`; “cand” = passing the interior filter; “excl” = additionally failing RSST condition 6; “valid” = the true RSST configuration class. Every “D-red” entry is 0. Per- (r, n) machine-readable counts in `results/theorem/fr_table/manifest_r*.json`; per-configuration records (adjacency, a , closure trace) in `configs_r*_n*.jsonl`.

r	interior range	raw	cand	excl	valid	D-red
8	1–4	436	18	6	12	0
9	1–4	666	35	17	18	0
10	1–5	13,338	221	127	94	0
11	1–6	293,555	1,631	1,028	603	0
12	1–6	509,288	4,092	3,004	1,088	0
13	1–7	12,389,976	36,392	27,888	8,504	0
total		13,207,259	42,389	32,070	10,319	0

RSST condition 6 is genuine structure and removes a lot. After normalising away the labelling convention (condition 4), a substantial fraction of accepted near-triangulations still fail condition 6 — some interior vertex meets the ring in more than two contiguous arcs. `reduce.c`’s own `ReadConf` rejects such graphs outright: they are not configurations in RSST’s technical sense at all. The exclusion rate rises from 33% at $r = 8$ to 77% at $r = 13$. Excluded graphs were nevertheless retained in the output with a D-reducibility verdict computed anyway, so that a D-reducible-but-not-RSST-legal graph would have been surfaced loudly. None was: every one of the 32,070 exclusions is non-D-reducible too, so the filter never changed a verdict, only the population count.

Condition 4 is a labelling convention, not structure. `ReadConf` requires each ring vertex’s neighbour list to start at a specific ring neighbour; nothing our checker computes depends on it, since a cyclic list denotes the same rotation system whatever entry it starts from. Naively serialising an RSST-relabelled configuration made `build/reduce_rsst` reject it with `ReadErr(4, ...)` even though the underlying embedded graph was fine. We added `rotate_ring_starts`, a mathematical no-op on the rotation system.

The $r = 13$ sweep is the expensive one: interior 6 ($n = 19$) and interior 7 ($n = 20$) were run 8-way sharded by residue class, $\approx 2,250$ s and $\approx 13,300$ s per shard respectively, giving 1,445 and 6,710 valid configurations and 0 D-reducible.

4.3 Triple verification

(1) Our checker vs. the compiled 1995 oracle. Up to 20 RSST-legal below-threshold configurations per ring were serialised and run through `build/reduce_rsst`, compiled from the pinned `reduce.c`: 70 sampled, 0 **verdict mismatches**, 6 oracle-inconclusive (`results/theorem/fr_table/crosscheck.json`).

(2) The six inconclusive cases, adjudicated by a third implementation. On 6 of the 70, the oracle recomputed a *different* $|\Phi(K)|$ than our checker supplied and aborted with ***** ERROR: DISCREPANCY IN NUMBER OF EXTENDING COLOURINGS ***** before printing a verdict. We wrote a third, from-scratch count of $|\Phi(K)|$ (`src/fourcolor/brute_force.py`) sharing no code path with the production checker — different edge indexing, different triangle enumera-

tion, no gauge fixing, no greedy variable ordering. It agrees with our checker in all 6 cases and disagrees with the oracle in all 6:

config	r	n	our a	reduce_rsst	brute force
fr-r8-n12-172	8	12	72	48	72
fr-r9-n13-269	9	13	147	94	147
fr-r10-n15-4669	10	15	367	230	367
fr-r10-n15-4678	10	15	365	243	365
fr-r10-n15-4613	10	15	408	250	408
fr-r11-n17-89446	11	17	995	532	995

Provenance note. Our own and the brute-force counts, and the ORACLE.INCONCLUSIVE status of exactly these six, are machine-readable in `results/theorem/fr_table/crosscheck.json` (which records `oracle_returncode:` 31 for each). The oracle’s *recomputed* counts in the middle column were transcribed from its `stdout` into `results/theorem/fr_table/THEOREM.md` §4.3 and are not retained in the JSON; they are the one set of numbers in this paper that is not re-derivable from a machine-readable artifact without re-running the oracle.

Our checker agrees with the oracle’s header a on all 3,455 hand-curated catalog configurations with zero mismatches (§5), so this is not a general defect. The evidence points to `reduce.c`’s `strip()` edge-numbering heuristic — a hand-optimised greedy ordering routine, documented in its own comments as chosen to shrink the search tree — being the outlier on a class of inputs (small, exhaustively generated, “generic” near-triangulations) its original validation against a hand-curated catalog never exercised. None of the six configurations’ D-reducibility verdict is in question: all are **False**, and the oracle never disputed the verdict, only the intermediate count, before erroring out. We report it because an honest cross-verification protocol surfaces every discrepancy it finds.

(3) Tightness witnesses. For each r , a catalog D-reducible configuration with exactly $f(r)$ interior vertices, verified both by our checker and by the oracle (`results/theorem/fr_table/tightness.witnesses.json`):

r	witness	n	interior	our a	oracle
8	D0003.conf (nl4ct D)	13	5	100	*** D-reducible ***
9	D0006.conf (nl4ct D)	14	5	211	*** D-reducible ***
10	D0015.conf (nl4ct D)	16	6	536	*** D-reducible ***
11	D0024.conf (nl4ct D)	18	7	1,363	*** D-reducible ***
12	D0143.conf (nl4ct D)	19	7	2,863	*** D-reducible ***
13	126.504366 (RSST 633)	21	8	6,954	—

Across the whole labelled corpus, the minimum interior size of a D-reducible configuration is 4, 4, 5, 5, 6, 7, 7, 8, 9, 10, 10 at $r = 6, \dots, 16$; the entries at $r = 8, \dots, 13$ are exactly $f(r)$, and the entries at $r = 14, 15, 16$ are upper bounds on f there for which we have no exhaustive lower-bound sweep.

4.4 The C-versus-D nuance

Theorem 4.1 is scoped to D-reducibility (no contract). C-reducibility is strictly weaker, and the RSST 633-configuration catalog itself contains configurations at ring sizes 8 and 11 with interior sizes *strictly below* $f(r)$ that are C-reducible but not D-reducible:

ring	catalog ident	n	interior	contract
8	2.126	12	4	$\{(1, 9), (3, 9), (5, 11), (7, 11)\}$
11	126.7566	17	6	$\{(15, 12)\}$
11	128.136	17	6	$\{(15, 12), (1, 12), (3, 12), (10, 15)\}$
11	7562.126	17	6	$\{(15, 12)\}$

So the corresponding minimum over all reducible (C-or-D) configurations is, at least at rings 8 and 11, strictly smaller, and the D-qualifier in Theorem 4.1 is not decoration. (No ring-9, 10 or 12 catalog member exists at interior $f(r) - 1$ in either catalog; that is a fact about which configurations the catalogs' authors happened to need, not evidence that C-reducible near-misses do not exist at other rings.)

4.5 A first out-of-sample test of the mass law

The $f(r)$ sweep produces, as a by-product, an exhaustive population of *below-threshold non-reducible* configurations with their exact a values. Comparing the maximum a over that population with the minimum a over the D-reducible population at the same ring gives a clean separation at every measurable ring:

r	max a (below threshold, non-red.)	min a (D-reducible)	gap
8		82	94 12
9		172	211 39
10		438	496 58
11		1,128	1,252 124
12		2,339	2,863 524
13		6,003	6,954 951

The $r = 13$ row is genuinely out of sample: the prediction (a gap) was made after the $r \leq 12$ data and before the $r = 13$ enumeration finished. This is the empirical content of the threshold half of the coloring-mass law (§6).

5 Independent verification of the 4CT corpus

Nothing in §3–§4 means anything unless our reducibility checker is right. This section reports what we did to establish that, and what we found while doing it. The checker `four-color.reduce.check` was written from the mathematics of [16] (the `reduce.tex` write-up, in particular its Theorem 3.2 machinery), *not* ported from `reduce.c`; so agreement with the C programs is genuine differential evidence rather than a tautology.

5.1 The three catalogs

Three points of precision that we have seen stated loosely elsewhere, including in our own earlier notes.

Table 3: Independent re-verification of the published catalogs. “Agree” means the computed $|\Phi(K)|$ and $|\text{consistent}|$ both match the published header fields *and* the configuration is confirmed reducible.

catalog	source	configs	agree	artifact
RSST 633	[16] <code>unavoidable.conf</code>	633	633	<code>results/differential-unavoidable-r14/report.jsonl</code>
Steinberger $\mathcal{U}_{2822}$	[19] <code>U.2822.conf</code>	2,822	2,822	<code>results/differential-U_2822-r16/report.jsonl</code>
2026 pool (pruned)	[11] <code>D/*.conf</code>	7,770	7,697	<code>results/p3/pool-verify/SUMMARY.json</code>

The RSST 633 are not all D-reducible. Exactly 249 are D-reducible; the other 384 are C-reducible via a nonempty published contract and are *not* D-reducible. The compiled oracle confirms the same split (249 ***** D-reducible *****, 384 ***** Contract confirmed ***** alongside 384 ***** Not D-reducible *****). What is verified for all 633 is the header fields and reducibility in the C-or-D sense. Ring sizes run 6–14, so no configuration was excluded by our ring bound.

Steinberger’s set is genuinely D-only. All 2,822 are D-reducible, all with header $b = 0$ (no contract), rings 6–16; 2,822/2,822 agree with the published headers, 0 disagreements, 13,103 s of checker CPU. This matters because §3.5’s citation chain rests on Steinberger’s degree-5/6 computation being against a correct set.

The 2026 pool is verified for 7,697 of 8,200. The verified run used the pruned pool of 7,770 (see below); of those, 7,697 have ring ≤ 16 and were verified D-reducible with 0 anomalies in 12.3 CPU-hours, and 73 at rings 17–18 are outside our engines’ ring budget and **remain unverified by us**. Together with the 430 removed configurations, the honest accounting is $8,200 = 7,697 \text{ verified} + 430 \text{ removed} + 73 \text{ unverified}$. Prior to this, the 2026 pool’s reducibility had never been independently verified: the authors’ own verification was external to the released code.

5.2 A legacy oracle that aborts, and what replaced it

We compiled two C oracles: `build/reduce_rsst` from [16]’s `reduce.c`, and `build/reduce_stein` from the Steinberger-extended version (needed because the unmodified 1995 code has `MAXRING = 14`).

`reduce_rsst` completes the 633 cleanly (Reducibility of 633 configurations verified). `reduce_stein` does *not* complete Steinberger’s 2,822: it processes 783 of them (of which 782 print ***** D-reducible *****) and then aborts with `SIGABRT` (exit 134), truncating mid-number. The crashing input is preserved as `build/crash783.conf`: a ring-16 configuration with $n = 28$ and $a = 139,480$. Our checker completes it in 227.7 s with $a = 139,480$, matching the published header. So the full-set verification of $\mathcal{U}_{2822}$ rests on our checker against the published headers, not on that binary; the binary corroborates the first 783 and then dies.

We also encountered the milder failure of §4: on six exhaustively generated near-triangulations, `reduce_rsst` recomputes $|\Phi(K)|$ incorrectly and self-aborts, and a third implementation adjudicates in our favour in all six. The two findings together are the practical reason the programme maintains *three* independent counts of $|\Phi(K)|$ rather than two.

5.3 Reproducing the 2026 pipeline

We reproduced the released C++ pipeline [11] end to end on the full 8,200-configuration pool (19,754 configurations after symmetry expansion) and the 84 rule shapes, in a Debug build with all assertions live. Total wall time 4 h 38 m. Every count matches the paper:

stage	result	wall
<code>combine_rules</code>	1,832 combined rules, 671 non-blocked	≈ 3 s
<code>enum_wheels</code> $d=7..11$	5,439/6,790/3,285/626/8 bad wheels	1 h 45 m
<code>enum_cartwheels</code>	10,094 bad cartwheels (9,366 at $d=7$, 728 at $d=8$; zero at $d=9, 10, 11$)	2 h 09 m
three gluing lemmas	all passed (5,365 / 1,618 / 298 cartwheels)	43 m

Separately we re-implemented the wheel-level stage in Python (`src/fourcolor/nl4ct.py`) and ran it differentially against the C++ on **all** 5,692,937 **wheel candidates** at hub degrees 7–11 (11,165 + 48,915 + 217,045 + 976,887 + 4,438,925): every degree reports `exact_match: true` with 0 false positives and 0 false negatives against the C++ ground truth. That validation is what licenses the symbolic charge form (1) used in §3.

We also replayed the RSST 1995 unavoidability argument: `discharge.c` from [15] against the five presentation files `present7` through `present11`, all five verified against the 633-configuration set (`results/discharge-rsst/replay.txt`).

5.4 Coverage measurement and a pool that is 430 configurations too large

Measuring which pool configurations the pipeline actually uses gives a sharply two-level picture. At the *wheel* level, only 85 of the 8,200 configurations ever act as a blocker anywhere across $d = 7..11$; the whole wheel-level covering structure has at most one concretization per wheel and at most 5 blockers per concretization. At the *cartwheel* (refinement) level, 7,422 distinct configurations appear as blockers over 2,253,185 logged blocking events — 90.5% of the pool, and that is a *lower* bound, because the instrumentation records only the first matching configuration per event. So the intuition that a small core carries the proof is right at stage 1 and wrong at stage 2.

We then asked which configurations are used by *no* stage. A first attempt tracked three roles (combined-rule blocking, wheel blocking, first-match cartwheel blocking) and found 673 unused; removing them made the pipeline fail — all three gluing checks aborted, and the bad-cartwheel count collapsed from 10,094 to 0. That failure is preserved (`results/p3/runs/batch1/`) and it is what discovered a *fourth* blocking role: the gluing lemmas also consult the pool (2,199 distinct glue-blockers). Tracking all four roles gives 430 **configurations with zero usage everywhere**. Removing exactly those and re-running the entire pipeline gives PASS with counts *identical to the full-pool proof in every field*: 1,832/671 combined rules, 5,439/6,790/3,285/626/8 wheels, 10,094 bad cartwheels, three gluing lemmas green (`results/p3/runs/batch1b/VERDICT.txt`, `counts.json`).

Proposition 5.1 (Certified). *The 2026 proof goes through verbatim with a pool of 7,770 configurations rather than 8,200, a reduction of 5.2%.*

This is a floor we reached, not a claimed minimum: no further prune batch was run, and the honest reading of the two-level coverage picture above is that further reduction is bounded by the cartwheel-level usage, which is nearly the whole pool.

Remark 5.2 (What was designed and not built). Our original plan (01-PROBLEM-STATEMENT.md) was a set-cover MILP minimising $|U|$, with an LP-dual lower bound. **No MILP was ever built or solved**; the design was replaced by batched prune-and-verify once the coverage measurement showed that the first-match cartwheel attribution makes a one-shot set cover unsound. We report this because the problem statement is in the repository and a reader might otherwise look for a solver artifact that does not exist. Theorem 3.1 is the useful residue of that line: an LP result, but an impossibility rather than an optimisation.

6 The coloring-mass law

6.1 Setting

For a configuration K with free completion S , ring size r , interior size k , $n = r + k$, write $a = a(K) = |\Phi(K)|$ and $W(r) := (2^r + 2)/6$. The organising conjecture of this section has two halves.

Conjecture 6.1 (Coloring-Mass Law). (T) Threshold. *There is an increasing $A(r)$ such that every D -reducible configuration of ring size r has $a \geq A(r)$; empirically $A(r) \approx 2.4^r/12.79$. (C) Cap. $a \leq B(r, k)$ with B increasing in k ; conjecturally the sharp cap $B(r, k) = W(r)(4/3)^{k-1}$.*

If both held, one would obtain analytic lower bounds on $f(r)$ for all r : the finite table of Theorem 4.1 would become an infinite structural law — *reducibility is a coloring-mass threshold phenomenon, and interior vertices supply the mass*.

The Bridge Lemma below is the bridge from colourings to the chromatic polynomial and underlies everything that follows.

Lemma 6.2 (Bridge Lemma; proven). *For every near-triangulation S of the disk, the number of tri-colourings of S is $P(S, 4)/4$; the S_3 action permuting the three edge colours is free on tri-colourings, so the number of tri-colouring classes is $P(S, 4)/24$; and consequently $a \leq P(S, 4)/24$.*

Proof sketch. Identify the three edge colours with the nonzero elements of the Klein group $V = \mathbb{Z}_2^2$ and label each edge uv of a proper 4-colouring c by $t(uv) = c(u) + c(v)$. Properness makes t a tri-colouring, and translation by V gives a 4-to-1 map. Conversely a tri-colouring is a V -valued 1-cochain summing to 0 around every internal face; since S is a triangulated disk the internal faces generate the cycle space, so t is exact and has exactly 4 potentials. For freeness: a transposition fixing t would force all edges to carry the one fixed colour, impossible in a triangle; a 3-cycle fixes no colour. Restriction to the ring sends each class of tri-colourings to one class of ring colourings, and $\Phi(K)$ is the image. $\square$

6.2 The wheel: exact closed form

Theorem 6.3 (Proven, with equality case). *If $k = 1$ then S is the wheel W_r and*

$$a = \frac{2^r + 2(-1)^r}{6} = \frac{P(W_r, 4)}{24}.$$

In particular the Bridge Lemma is tight at every wheel; $a = W(r)$ for even r and $a = \lfloor W(r) \rfloor$ for odd r .

Proof. $k = 1$ forces the internal faces to be $hx_i x_{i+1}$, so $S = W_r$, and interior degree ≥ 5 forces $r \geq 5$. Writing e_i for the rim edge colour and s_i for the spoke colour, the face condition says $\{s_i, e_i, s_{i+1}\}$ is all three colours, so $s_i \neq s_{i+1}$ and e_i is determined; hence $\#\text{TriCol}(W_r) = P(C_r, 3) = 2^r + 2(-1)^r$. Let ρ be the restriction to the rim. Given (e_i) and s_1 , the face condition forces $s_{i+1} = \tau_{e_i}(s_i)$ where τ_c transposes the two colours $\neq c$; so the spoke vector is determined by s_1 , which must avoid e_r and e_1 . If $e_i \neq e_{i+1}$ for some i then s_{i+1} avoids two distinct colours and the fibre has size ≤ 1 . If (e_i) is monochromatic the spokes alternate between the two colours $\neq c$, which closes up iff r is even, giving a fibre of size exactly 2 (and empty for odd r). An element of S_3 fixes (e_i) iff it fixes every e_i ; a 3-cycle fixes no colour and a transposition fixes one, so the rim colourings with nontrivial stabiliser are exactly the three monochromatic ones. For odd r no monochromatic rim colouring is in the image, every fibre is a singleton, S_3 acts freely, and $a = |\text{TriCol}|/6 = (2^r - 2)/6$. For even r the six tri-colourings above the three monochromatic rim colourings form a single free S_3 -orbit — one class — while the three rim colourings likewise form one element of Φ , so $a = (|\text{TriCol}| - 6)/6 + 1 = (2^r + 2)/6$. $\square$

The wheel value is $P(C_r, 3)/6$, the number of proper 3-colourings of a cycle modulo colour symmetry. Everything in the sharp cap's constant is that one classical number. The theorem was adversarially re-verified four independent ways for $r = 5..14$ — our production checker, the brute-force tie-breaker, a from-scratch enumeration of spoke sequences with explicit S_3 canonicalisation, and the compiled 1995 RSST oracle (which covers $r \leq 12$ and hard-errors at $r \geq 13$ on a compile-time array cap, a build limit rather than a disagreement).

6.3 The star-first cap

Theorem 6.4 (Proven). *For every configuration K with $k \geq 1$,*

$$a \leq \min_{h \text{ interior}} \frac{(2^d + 2(-1)^d) 2^{n-d-1}}{6}, \quad d = \deg h,$$

and since every interior degree is ≥ 5 , uniformly

$$a \leq 11 \cdot 2^{r+k-7} = \frac{11}{16} 2^{r+k-3}.$$

The bound is exactly attained at every wheel.

Proof sketch. Let h be interior of degree d and $L = N(h)$ its link, a cycle (not necessarily induced — a chord only removes colourings, so $P(C_d, 3)$ remains valid). Order the vertices $h, x_1, \dots, x_d, w_1, \dots, w_{n-d-1}$ so that every w_j has two predecessors adjacent to each other: grow greedily from the closed star of h by attaching any triangle sharing an edge with the built region. Such a triangle introduces at most one new vertex, adjacent to both ends of the shared edge. The greedy exhausts the rest because the dual graph of S is connected and, for any triangle T not incident with h , the last triangle on a dual path from T to a star triangle shares a *link* edge $x_i x_{i+1}$ (it cannot share hx_i , which lies only in star triangles). Counting along the order: h has 4 choices, L has at most $P(C_d, 3) = 2^d + 2(-1)^d$, and each w_j has at most 2. Divide by 24 using Lemma 6.2. For the uniform form, $2^{6-d}(2^d + 2(-1)^d)/6$ equals 10, 11, 10.5, 10.75, 10.625, $\dots \rightarrow 32/3$ for $d = 5, 6, 7, 8, 9, \dots$, with maximum 11 at $d = 6$. $\square$

Note the counter-intuitive shape: *odd* interior degrees give the better bound, since $P(C_d, 3) = 2^d - 2$ for odd d . Consequently the worst case for the star-first argument is a

configuration whose interior degrees are all even and ≥ 6 ; and if S has an interior vertex of degree exactly 5 (which 58,977 of our 59,142 corpus configurations do) then $a \leq 5 \cdot 2^{r+k-6}$. The 165 exceptions are all-degree- ≥ 6 interiors including the six wheels, which is exactly why the unconditional constant must be 11 rather than 10.

Theorem 6.4 improves the constant of the previously proven weak cap $a \leq 2^{r+k-3}$ by a factor 11/16 and is exactly tight at $k = 1$; but its k -dependence is still 2^k , so it moves the offset of the implied $f(r)$ bound and not its slope: $f(r) \geq \lceil 0.26303r - 0.1367 \rceil$ instead of $\lceil 0.26303r - 0.6768 \rceil$. That is +1 interior vertex at 6 of 10 rings, real, small, and free.

6.4 $\gamma = 3/2$ at $k = 2$, and why the induction stops

Order the interior graph by BFS from a vertex h_1 of maximum degree d_1 . Colour h_1 and its link; for $j \geq 2$, split $\text{link}(h_j)$ into maximal arcs of not-yet-coloured vertices, each an at-most-3-colour path with both ends constrained, so an arc of length t admits at most $M(t) = 2, 3, 6, 11, 22, 43, \dots$ colourings. Writing $g(t) = M(t)/2^t \in \{1, 3/4, 3/4, 11/16, \dots\} \rightarrow 2/3$ gives the per-configuration certificate

$$a \leq (1 + 2(-1)^{d_1} 2^{-d_1}) \cdot \frac{2}{3} \cdot 2^{n-3} \cdot \prod_{\text{arcs}} g(t), \quad (3)$$

which reduces exactly to Theorem 6.3 at $k = 1$.

Theorem 6.5 (Proven, $k = 2$ only). *For $k = 2$, $a \leq (1 + 2(-1)^{d_1} 2^{-d_1}) \frac{2^r}{6} \cdot \frac{3}{2} \leq \frac{33}{32} \cdot \frac{2^r}{6} \cdot \frac{3}{2}$.*

Proof. Every edge of S joining two interior vertices lies in exactly two triangles, hence has at least two common neighbours; a third common neighbour z would make $h_1 h_2 z$ a separating triangle, which at $k = 2$ is impossible since the only vertices are h_1, h_2 and the ring, and the ring lies outside. So the coloured part of $\text{link}(h_2)$ at step 2 is exactly $\{h_1, x, y\}$, the complementary arc is a single arc of $\deg(h_2) - 3 \geq 2$ vertices, and $\prod g = g(\deg(h_2) - 3) \leq 3/4$. Substituting in (3) with $2^2 \cdot (3/4) = 3$ gives the claim. $\square$

The “exactly two common neighbours” fact was machine-checked on all 610,846 interior-interior edges of the corpus with zero exceptions, but it is only *proved* at $k = 2$.

Why the induction stops, precisely. The natural generalisation — hypothesis (H2), that every h_j with $j \geq 2$ contributes at least one arc of length ≥ 2 — would give $\gamma = 3/2$ for all k . **(H2) is false on the corpus.** Of the 372,008 BFS steps with $j \geq 2$, 20,379 (5.48%) contribute no arc at all and 119,705 (32.18%) contribute only length-1 arcs, which gain nothing since $g(1) = 1$; the length-1-only fraction climbs from $\approx 3\%$ at $k = 3$ to $\approx 60\%$ at $k = 16$. The realised effective base $\gamma_{\text{prov}} = 2(\prod g)^{1/(k-1)}$ has maximum 1.5000, 1.7321, 1.8171, 1.8612, 1.8882, 1.9064, 1.9195 at $k = 2, \dots, 8$ and 1.937082 overall. So this scheme yields *no uniform $\gamma < 2$ beyond $k = 2$* . The value $1.7321 = 2\sqrt{3/4}$ at $k = 3$ is the signature of “one gaining step out of two”.

Moreover $g(t) \geq 2/3$ for every t , so *no argument of this shape can ever give $\gamma < 4/3$* : the empirical near-tightness of $4/3$ is not an accident, it is the scheme’s infimum, attained only for arbitrarily long arcs.

6.5 The obstruction theorem

Theorem 6.6 (Proven; explicit 10-vertex witness). *Let S_0 be the free completion of the corpus configuration **gen-r8-n10-2** ($r = 8$, $k = 2$, both interior vertices of degree 6). Then exactly*

$$P(S_0, 4) = 1464, \quad \frac{P(S_0, 4)}{24} = 61, \quad a(S_0) = 55, \quad W(8) \cdot \frac{4}{3} = \frac{172}{3} = 57.33 \dots$$

Hence $P(S_0, 4)/24 > W(8) \cdot (4/3) > a(S_0)$, so **no upper bound on a that factors through the Bridge Lemma — i.e. any proof of the form $a \leq P(S, 4)/24 \leq B(r, k)$ — can prove the sharp cap**, and more precisely no such proof can establish $a \leq W(r)\gamma^{k-1}$ for any $\gamma < 61/43 = 1.4186 \dots$

Over the whole corpus the smallest γ for which the $P/24$ form survives is $\max \gamma_P = 1.4186047$ at exactly this record, against $\max \gamma_a = 1.3224310$ at **fr-r13-n15-4**; and the P -form of the sharp cap is outright *killed*, failing on 1,960 of 59,142 records (3.3%), worst ratio 1.3707.

The Bridge-Lemma fibre $P(S, 4)/(24a)$ — two colour-inequivalent tri-colourings sharing a ring restriction — is exactly 1 at $k = 1$ and grows like $1.05\text{--}1.08^k$. It is that factor the sharp cap silently uses. A proof of the sharp cap must therefore contain a mechanism that sees *ring restrictions*, not just tri-colourings.

The per-step barrier. There is a second, independent reason $4/3$ is hard. For any order in which each vertex has $b_i \geq 2$ earlier neighbours, the edge count $m = 3n - r - 3$ gives the exact identity $\sum_{i \geq 4} (b_i - 2) = k$: the “excess adjacency” budget is exactly one closing per interior vertex. A closing step with earlier neighbours forming a path $u-x-w$ has 2 colours available if $\varphi(u) = \varphi(w)$ and 1 otherwise, i.e. an average factor $1 + \Pr[\varphi(u) = \varphi(w)]$. So the cap base is exactly $\gamma = 1 + \Pr$, and $\gamma = 4/3$ means $\Pr \leq 1/3$, i.e. u and w *exactly uncorrelated* given their common neighbour. That is a sharp mean-field statement at a critical point: the $q = 4$ antiferromagnetic Potts model on the triangular lattice is critical [13], correlations do not decay exponentially, and the standard correlation-decay techniques [17] need $q \geq 11$ there. Indeed $4/3 = (q-2)^2/(q-1)$ at $q = 4$ is Shrock–Tsai’s *lower* bound for the triangular-lattice entropy [18] — a floor, not a ceiling.

We measured the distribution directly. Over a stratified sample of 1,315 configurations covering every occupied (r, k) cell, the exact step factors $f_i = P(G_i, 4)/P(G_{i-1}, 4)$ along the star-first order are:

b_i	mean f_i	min	max	steps
2	2.0000	2.0000	2.0000	14,886
3	1.3155	1.1754	1.6479	6,322
4	0.8583	0.7273	1.0933	1,165
≥ 5	0.5138	0.2148	0.6576	565

The $b_i = 3$ row is exactly $1 + \Pr[\varphi(u) = \varphi(w)]$. Its mean 1.3155 sits just below the mean-field $4/3$ — which is why the sharp cap fits so well on average — but its *maximum* is 1.6479, i.e. $\Pr = 0.648 > 1/2$. So no per-closing-step bound can give $\gamma = 3/2$, let alone $4/3$: individual closings are demonstrably worse than either target, and any proof must average over them.

6.6 The sharp cap survives where the Bridge Lemma does not

Baxter’s ground-state entropy of the 4-state antiferromagnetic Potts model on the triangular lattice is ≈ 1.4610 per site [3], greater than $4/3$. A disk cut from the triangular lattice has all interior degrees $6 \geq 5$ and an induced boundary cycle, so it lies in our class — and it predicts that the P -form of the cap must fail badly in the bulk. It does: on exact $P(S, 4)$ computations for hexagonal disks up to $R = 7$ ($n = 169$) and corner-truncated triangular disks up to $m = 16$, the ratio $(P/24)/\text{cap}$ reaches 1.1×10^4 , with fitted per-site entropy $1.541/1.563$.

But $a = |\Phi|$ itself, computed directly on the same disks by enumerating proper 4-colourings, taking Klein edge labels, restricting to the boundary cycle and canonicalising under S_3 (self-validated 12/12 against the repository’s stored a), **never violates the sharp cap, and the margin improves as the disk grows**:

disk	r	k	a	cap	a/cap	γ_a
HEX $R=1$ ($= W_6$)	6	1	11	11.0	1.0000	—
TRI $m=4$	9	3	140	152.3	0.9193	1.278
TRI $m=5$	12	6	2,263	2,878.2	0.7863	1.271
HEX $R=2$	12	7	2,756	3,837.5	0.7182	1.262
TRI $m=6$	15	10	41,971	72,740.1	0.5770	1.255
TRI $m=7$	18	15	853,124	2,452,078	0.3479	1.2365

The Bridge-Lemma fibre on these disks is 1.21, 1.65, 1.93, 2.81, 6.30 at $k = 3, 6, 7, 10, 15$: it grows fast enough to absorb the entire bulk excess and then some. These disks were constructed adversarially to break the sharp cap and did not. What is dead is the route, not the statement.

6.7 The threshold half, empirically

Mining the corpus for the exact frontier gives, per ring, the minimum a over D-reducible configurations:

r	#D-red	min a	k^*	witness
6	28	16	4	0.7322 (RSST 633)
7	37	39	4	2.122 (RSST 633)
8	1,435	94	5	gen-r8-n13-1494 (generated)
9	2,168	211	5	2.7566 (RSST 633)
10	1,891	496	6	122.140 (RSST 633)
11	209	1,252	7	122.1028 (RSST 633)
12	767	2,863	7	126.453966 (RSST 633)
13	1,961	6,954	8	126.504366 (RSST 633)
14	2,654	17,440	9	2039 (Steinberger)
15	1,562	42,957	10	2420 (Steinberger)
16	457	100,454	10	D3848 (nl4ct pool)

Two structural observations. First, **at every one of the eleven rings the minimum- a witness sits at the minimum possible interior count** — on rings 8–13 that is exactly the $f(r)$ table. The coloring-mass frontier and the interior-count frontier are attained by the *same* configurations; the mass law is not an independent phenomenon layered on top of Theorem 4.1. Second, the residual $\min a(r)/2.4^r$ is flat: 0.07823 to 0.08540 over eleven rings, a spread of

Table 4: $f(r)$ implied by pairing a threshold with a cap. “ $\gamma = 4/3$ ” is the sharp cap of Conjecture 6.1, not proven.

r	8	9	10	11	12	13	14	15	16	20
true $f(r)$	5	5	6	7	7	8	?	?	?	?
weak cap 2^{r+k-3} (proven)	2	2	2	3	3	3	4	4	4	5
Thm 6.4 $11 \cdot 2^{r+k-7}$ (proven)	2	3	3	3	4	4	4	4	5	6
$\gamma = 3/2$ (proven at $k = 2$ only)	3	4	4	5	5	5	6	6	7	9
$\gamma = 4/3$ (the target)	4	5	5	6	6	7	8	8	9	12

1.0916 across a range in which 2.4^r itself grows by a factor 6,340. Sweeping the base in steps of 0.005 over $[2.30, 2.50]$, the spread-minimising base is $\beta = 2.400$ exactly. There is no detectable polynomial correction: the law is $A(r) = c\beta^r$ with a single constant.

The tightest surviving pure-exponential form is

$$(T1) \quad K \text{ D-reducible} \implies a \geq 2.4^r/12.79,$$

with $a \geq 2.4^r/12.78$ KILLED. Its per-ring margins lie in $[1.001, 1.092]$; it is the only candidate near-tight at both ends and in the middle at once. **(T1) is corpus-supported only.**

Table 4 shows the payoff. Pairing (T1) with the sharp cap gives $f(r) \geq \lceil 0.63375r - 1.6306 \rceil$, i.e. 4, 5, 5, 6, 6, 7 against the true 5, 5, 6, 7, 7, 8: **off by at most one at every ring, from two inequalities**. That is an analytic near-reproduction of the entire $f(r)$ theorem, valid for all r . With the *proven* cap the implied slope is only 0.263, so the entire remaining factor of 2.4 in f -slope lives in the cap, not the threshold.

Remark 6.7 (The structural principle). Under a cap $B(r, k) = C \cdot 2^r \gamma^k$, the implied f -slope is $(\log_2 \beta - 1)/\log_2 \gamma - 0.263$ at $\gamma = 2$, 0.450 at $\gamma = 3/2$, 0.634 at $\gamma = 4/3$. *The entire source of f -growth is the excess of the threshold base β over 2*; a law with $\beta \leq 2$ proves nothing about f . This kills the whole wheel-anchored family of threshold candidates as a route to f , since their r -dependence 2^r cancels against the cap’s.

Remark 6.8 (Honest caveat on the pairing). (T1)’s implied slope 0.634 slightly *exceeds* the empirical f -slope (≈ 0.60 , the endpoint slope $(8,5) \rightarrow (13,8)$; least squares over the six points gives 0.629, which is not significantly below 0.634). Extrapolated at the endpoint slope the implied bound would overtake the truth around $r \approx 54$; at the least-squares slope, not until $r \approx 263$. So at least one of $\{(T1), \text{the sharp cap}, \text{the } 0.60 \text{ slope estimate}\}$ must weaken before then.

Remark 6.9 (Where the corpus does and does not constrain). Only rings 6–10 have machine-generated D-reducibles; at rings 11–16 the entire D-reducible population is the union of the three published catalogs, and at rings 15–16 there is not even a non-reducible negative population. A frontier read off rings 14–16 measures what catalog authors chose to publish. We tested this: re-running the winners restricted to non-catalog sources (51,067 records evaluated, of which 5,478 are the D-reducible witnesses that can bind; effectively rings ≤ 10) leaves the generated-only frontier $\{6:19, 7:39, 8:94, 9:228, 10:516\}$ versus the all-source $\{6:16, 7:39, 8:94, 9:211, 10:496\}$, tightening the $\beta = 2.4$ constant by only 3.9%; and $a \geq 2.4^r/12$ is killed by the generated population independently of any catalog. So the frontier is not a catalog artifact on the rings where we can check — but that check covers only $r \leq 10$.

7 Methods: the verification architecture

The programme’s operating assumption is that on this material a single implementation’s output is worthless, and that the interesting failures are the ones a plausible-looking pipeline produces silently. Five mechanisms carry the load.

(M1) Two (and where needed three) independent implementations. Our reducibility checker is written from the published mathematics, not ported from the released C. Where the two disagree, a *third* implementation with no shared code path adjudicates (`src/fourcolor/brute_force.py`: different edge indexing, different triangle enumeration, no gauge fixing, no greedy variable ordering). The same discipline is applied to the discharging side: the Python port of `always_apply/never_apply/blocking` was differentially validated against the C++ on all 5,692,937 wheel candidates before being trusted, and re-validated on all 10,094 published leaves before being used in §3. In both places the differential test found real defects — in the legacy C in one direction (§5) and in our own serializer in the other (§4.2).

(M2) Adversarial mutation testing. A candidate statement that holds on a pool has, on the evidence of §9, essentially no chance of being true. `tools/stress_test_theorem.py` attacks a candidate by generating mutants of pooled configurations under a single diagonal edge flip (`src/fourcolor/mutate.py`), filtering each mutant through a structural legality gate (a port of RSST’s own `ReadConf` conditions), then through the candidate rule on recomputed features, then through a canonical-key novelty check; survivors are labelled by the reducibility checker and every false positive is re-adjudicated by the compiled C oracle. The one time we ran this on genuinely novel candidates it killed all three of them (§9).

(M3) The lemma harness: test before proving. Any candidate statement can be machine-checked against the entire labelled corpus of 59,142 canonically deduplicated configurations (rings 6–16, 13,169 of them D-reducible) in about 0.2 s for a scalar statement:

```
tools/test_lemma.py --check "rec.d_reducible implies rec.a >= 2.4**rec.r/12.79"
```

Each run appends a receipt to `results/mass-law/lemma_log.jsonl`, content-hashed against the corpus signature `a2b1feba331bf2962b2b` — a blake2b digest over (`canonical_hash`, `r`, `n`, `a`, `b`, `d_reducible`) for records in canonical-hash order, so a receipt identifies exactly which configurations with which labels a verdict was taken over. Every corpus-supported claim in §6 carries such a receipt. The harness’s verdict vocabulary is deliberately restricted: `HOLDS` means “survived the corpus”, never “proved”. Of the 87 logged receipts, 66 are `HOLDS` and 21 are `KILLED`; the killed ones are as much of the output as the surviving ones.

(M4) Exact-arithmetic certificates with re-derivation. Where a result is a computation, we ask it to emit an object that an independent, simpler program can check. `tools/verify_farkas.py` is the archetype: solver-free, `Fractions.Fraction` throughout, and — the part that matters — it does not trust the certificate’s stored coefficients, re-deriving every row from its source artifact and failing loudly on mismatch. Corrupting one coefficient by hand is rejected with an exact claimed-versus-re-derived diff. The corresponding discipline

on the enumeration side is that generation completeness is delegated to a pinned canonical-construction-path generator [5] rather than to our own code, and legality to a port of the original authors’ own parser.

(M5) Adversarial review of every claim before it is written down. Each major result was handed to a reviewer prompted to *refute* it and to default to “refuted” when unsure, with independent reimplementations required. This produced, among other things: the discovery that the pool-independence claim in Lemma 3.3 was false (fixed); the code-necessary versus math-necessary distinction that became §3.6; the coarse-cartwheel gap in the low-degree rows (closed exactly, §3.5); the demotion of a claimed $\gamma = 3/2$ cap for all k to Theorem 6.5 at $k = 2$ only, after the reviewer showed the enabling hypothesis fails on 90.5% of the corpus; and two false provenance sentences about [11] that had survived several drafts. We record this as a method because in every one of those cases the error was in prose or scope rather than in a computation, and no amount of differential testing would have caught it.

Test suite and pinning. 23 test files, 295 discovered test cases, run by `make test`; at the time of writing 293 pass and 2 fail. Both failures are a data-drift guard firing correctly rather than a defect: the ring-8 trace corpus grew from 1,162 to 1,736 records after the ring-8 model was trained, so `verify_split_against_metrics` refuses to reconstruct that model’s train/validation split against the enlarged data. Ring 8 carries none of the results reported in §8, which are at rings 9 and 10. All third-party inputs are pinned: four arXiv e-prints with recorded SHA-256 digests (`third_party/CHECKSUMS.sha256`), three git commits for the released 2026 artifacts (`third_party/GIT_PINS.txt`), and `plantri` 5.5 with the digests of both binary and source.

8 Machine learning: what was tried, and what it gave

This section is an accounting, not an advertisement. None of the theorems in this paper were discovered by a neural model. They came from exhaustive enumeration, LP duality, and machine-checked mining over finite catalogs, as described in the preceding sections. We nonetheless ran a substantial interpretability programme alongside that work, on the hypothesis that a small model trained to predict D-reducibility might expose a human-statable sufficient condition that the enumeration was not organized to find. We report what that programme gave: one modest positive result, one methodological finding about the fragility of “mechanism” as a concept, and several negative results we think are worth recording so that others do not have to rediscover them. We write in the register of a referee report on our own work.

8.1 Design and motivation

The programme deliberately transplanted the template of Huang, Jackson, and Lee’s mechanistic-interpretability study of the zeta map [10]: train a small transformer on a well-defined mathematical computation, probe its internal representations, and try to extract a human-readable mechanism. That template sits inside the broader agenda of using machine learning to guide mathematical intuition [8]. The target computation here was D-reducibility itself: given a configuration, predict the Kempe-closure trace produced by the D-reduction search and the final verdict (reducible or not). The rationale was simple. If a small model can

predict reducibility reliably, whatever internal feature it relies on is a candidate for a sufficient condition a human could then state and verify independently – the same logic that motivated the zeta-map study’s search for a legible mechanism.

We built two model generations.

D1. An encoder-decoder transformer ($d_{\text{model}} = 192$, 4 heads, feed-forward width 384, 2 encoder layers, 2 decoder layers, dropout 0, 1,543,725 parameters), trained seq2seq from a canonical token sequence encoding a configuration to a target sequence of closure-trace count buckets followed by a verdict token. Training ran 120 epochs at learning rate 10^{-3} , batch size 64, seed 0, for 10,392 s on an Apple MPS device. The corpus, `data/d1.corpus.jsonl`, held 14,036 deduplicated records spanning ring sizes 6–16 (633 from RSST, 2,614 from Steinberger, 5,961 generated, 4,828 drawn from the 2026 pool); after excluding records with degenerate traces or out-of-range length, 6,559 remained, split 5,904 train / 655 validation. Artifacts: `checkpoints/d1.full.v1.pt`, `results/d1/train_metrics_full.v1.json`, `results/d1/corpus_stats.json`.

D1-v2. A second generation of per-ring specialists at $r = 8, 9, 10$, sharing a 2-layer encoder ($d_{\text{model}} = 256$, 4 heads, feed-forward width 512, masked-mean-pool summary) with two alternative heads. A “dynamics” head iterates a weight-tied GRUCell over the per-ring survivor-set bitmap with per-round supervision (3.26M parameters at $r = 9$, 6.64M at $r = 10$); a “verdict-only” CONTROL head reads the verdict directly off the encoder summary with no recurrence (1.19M parameters), and exists specifically to test whether the dynamics head’s recurrent machinery contributes anything the summary vector does not already contain. Data lived in `data/v2/traces_r8.jsonl`, `data/v2/traces_r9.jsonl`, and `data/v2/traces_r10.jsonl`, carrying full per-round survivor sets; the corpora held 1,162 / 1,446 / 1,746 configurations at $r = 8, 9, 10$, with ring-colouring code-index sizes 1,094 / 3,281 / 9,842 respectively. Artifacts: `results/d1v2/r9/`, `results/d1v2/r10/`, `results/d1v2/r10v2/`.

8.2 Results

D1 reaches verdict accuracy 0.9298 on its 655-example validation split, against a majority-class baseline of 0.5603 – a real signal, well above chance. Full-sequence exact match (verdict token plus the entire closure-trace bucket sequence) is a much weaker 0.2397: the model gets the answer right far more often than it gets the reasoning trace right.

We then asked what the encoder actually represents, using a probe ladder recorded in `results/d1/interp/report.md` (ridge logistic probes on standardized inputs, probe-test $n = 131$, 95% bootstrap confidence intervals with `n.boot= 2000`). A 16-feature hand-coded shallow baseline – counts and simple graph statistics of the configuration, no learned representation involved – already reaches 0.8779 [0.8168, 0.9313]. The best pure-encoder representation is `encoder_layer0_mean`, the *pre-attention* embedding bag, at 0.9008 [0.8473, 0.9466]: a delta of +0.0229 over the shallow baseline, but with overlapping confidence intervals, so not a distinguishable improvement. The post-attention layers are worse, not better: layers 1 and 2 both give 0.8855, despite fitting the training set considerably better than layer 0 (train accuracy 0.9599 and 0.9866 respectively, against 0.8817 for the pre-attention bag). Attention is improving the model’s memorization of the training distribution without improving the generalizing signal a downstream probe can extract. Augmenting the shallow features with hand-derived structural candidates reaches 0.9237 [0.8779, 0.9695] – that is, hand features beat the encoder’s best representation outright. We state this plainly as the honest summary of D1: the trans-

former’s internal representation of reducibility is, on this evidence, no better than what a referee could write down with a pencil, and in places worse.

A second question was whether the decoder accumulates verdict evidence as it emits the closure trace, the way one might hope a legible mechanism would. We probed decoder hidden states at each greedy decoding step $t = 0, \dots, 39$. Verdict-probe accuracy is 0.8473 at $t = 0$, before a single trace token has been produced, and the curve stays essentially flat thereafter: the mean over $t = 0\text{--}2$ is 0.8702, the mean over $t = 35\text{--}39$ is 0.8656, and every later confidence interval overlaps the one at $t = 0$. The closure-trace tokens ride along with a verdict decision already implicit at the start; they do not build toward one.

8.3 The boundary result

The results above concern accuracy on the full validation distribution, which is dominated by easy cases. We pre-registered a harder “boundary” stratum, flagged by `tools/dlv2_datagen.py::boundary_flag`, to test whether any representation carries information beyond the shallow features specifically on the cases an easy classifier would get wrong: a configuration is boundary if it is non-reducible with $0 < n_{\text{consistent}} \leq 50$, or reducible with rounds ≥ 6 . We pre-registered the decision rule before looking at boundary results: a representation beats the shallow baseline only if its boundary 95% bootstrap confidence interval does not overlap the shallow baseline’s and lies strictly above it.

The first pass, at rings 9 and 10 on the smaller corpus, was a clean null: 0 of the 7 non-baseline representations at $r = 9$ and 0 of 7 at $r = 10$ cleared the rule, and the boundary strata in the probe-test splits were tiny (7 and 11 examples respectively) – underpowered by construction.

The second pass used an enlarged ring-10 corpus (3,269 configurations; probe-test $n = 98$, of which 21 are boundary) and gave a real result. Table 5 summarizes it.

Representation	Overall accuracy	Boundary accuracy [95% CI]
shallow	0.7551	0.5238 [0.3321, 0.7143]
shallow_plus_structural	0.8571	0.7619 [0.5714, 0.9048]
control_encoder_summary	0.9082	0.9048 [0.7619, 1.0000]
dynamics_encoder_summary	–	0.9048
dynamics_h0	–	0.9048
dynamics_h2	–	0.9048

Table 5: Ring-10 boundary probe results, second pass ($n = 98$ overall, 21 boundary). `control_encoder_summary` and three dynamics-model representations clear the pre-registered rule against the shallow baseline.

`control_encoder_summary` – the 256-dimensional encoder summary feeding the verdict-only control head, with no recurrent dynamics involved at all – clears the pre-registered rule against the plain shallow baseline, as do three dynamics-model representations. This is the one genuinely positive result in the programme: some encoder representation carries boundary-relevant information the 16 hand-picked shallow features do not.

We caveat it on three counts. First, $n = 21$ boundary examples is a small stratum; the confidence intervals in Table 5 are wide. Second, against the also-hand-engineered `shal-`

`low_plus_structural` baseline the comparison is never confidence-interval-decisive at any seed we tried (0.9048 versus 0.7619, intervals overlap); the result holds against the naive 16-feature baseline, not against a reasonably strengthened one. Third, we note for the record that the source report’s own concluding prose (`results/d1v2/v2run/interrogation.md`) contradicts its table: it recycles wording from the earlier null run at rings 9 and 10. We report the table, not the prose.

8.4 Extraction: hand features recover most, but not all, of the advantage

Given that some encoder representation beats the shallow baseline on boundary cases, we tried to extract what it was finding, in the spirit of turning a learned feature into a stated one. We hand-coded 14 arrangement features of H_5 , the subgraph induced on interior vertices of degree exactly 5 (`src/fourcolor/d1_features.py`): number of connected components, largest component size, number of isolated vertices, edge count, density, maximum degree, existence of a length-3 path, number of vertices adjacent to the ring, minimum and mean component-to-ring distance, minimum and mean inter-component distance, the count of degree-5 vertices with at least 3 degree-5 neighbours, and the existence of a diamond in H_5 . Appending these 14 to the 24-dimensional `shallow_plus_structural` vector (16 shallow features plus 8 structural ones) gives a 38-dimensional `shallow_plus_arrangement` vector. Table 6 reports boundary accuracy across four seeds, drawn from `results/d1v2/extraction/report.md`.

Seed	n_{boundary}	shallow	+structural	+arrangement [95% CI]	control.encoder
0	21	0.5238	0.7619	0.9524 [0.8571, 1.0000]	0.9048
1	23	0.4348	0.7826	0.9130 [0.7826, 1.0000]	0.7826
2	24	0.5417	0.7083	0.8750 [0.7083, 1.0000]	0.7500
3	25	0.5200	0.7600	0.7600 [0.5990, 0.9200]	0.9200

Table 6: Boundary accuracy of the hand-coded arrangement-feature extraction against the plain shallow baseline, the structural baseline, and the encoder control, across four seeds.

The extraction succeeds on three of four folds: at seeds 0, 1, and 2 the arrangement features push boundary accuracy well above `shallow_plus_structural`, and in all three above the encoder control as well (0.9524 vs 0.9048, 0.9130 vs 0.7826, 0.8750 vs 0.7500). It fails on seed 3: the arrangement features add exactly nothing over the structural baseline there, an identical point estimate of 0.7600 to 0.7600, while the encoder control reaches 0.9200. We do not know why seed 3 is different; we simply report that it is.

The boundary result of Table 5 itself replicates across these same four seeds in direction but not always in confidence-interval strength: the control encoder beats the plain shallow baseline by +0.38, +0.35, +0.21, and +0.40 at seeds 0–3 respectively, so the sign is consistent 4 of 4. But the confidence-interval-decisive version of the comparison – the pre-registered rule – holds only at seeds 0 and 3, i.e. 2 of 4. The headline boundary result reported in of §8.3 was therefore somewhat fortunate in its seed choice; a differently seeded first pass could plausibly have reported an inconclusive result instead. Artifacts: `results/d1v2/extraction/report.md`, `results/d1v2/replication/report.md`.

We also performed a small error-set digestion at seed 0 (`results/d1v2/digestion/cases.md`). Of the 21 boundary probe-test examples, the

encoder control is right and the structural baseline is wrong on 4; the reverse holds on 1; both are right on 15; both are wrong on 1. All 4 configurations where the encoder wins have $n = 18$, are D-reducible, and have closure round counts 6, 6, 7, 6. Their mean count of degree-5 diamonds is 2.75, against 0.88 for the rest of the validation boundary set and 0.97 corpus-wide – roughly a factor of three – with every other feature we measured flat across the two groups. We decline to run a significance test at $n = 4$; the sample is too small to support one honestly.

We flag one honest gap here. The automated per-configuration check that the arrangement features specifically fix these four cases did not actually run: `tools/d1v2_extraction_test.py` guards its `error_set_check` behind `hasattr(interro, "probe_predictions")`, which is `False` for the interrogation object actually produced, so the check is silently skipped and `error_set_check` is empty at every seed. The causal link we would like to draw – diamond count explains the four flipped cases – is therefore suggestive, resting on the aggregate statistic above, and not established at the level of individual predictions.

8.5 Mechanism is training-contingent: a replication finding

This subsection reports a companion experiment run in a separate repository, `/Users/yugendren/experiments/zeta_map_interp`, not in the codebase for this paper. Its subject is the published zeta-map result we took as our template, [10], and the question is whether its reported mechanism replicates under a faithful reimplementation.

We reimplemented the published architecture (1 encoder layer, 1 decoder layer, 1 attention head, $d_{\text{model}} = 128$, feed-forward width 256, post-norm, learned positional embeddings, weight-tied output projection): 337,029 parameters against the paper’s reported 339,716. Training reached the same headline accuracy the paper reports: 0.9950 greedy exact-match on a held-out instances of size $n = 13$. We then reran each of the paper’s three reported mechanistic findings against our own checkpoint.

- *Causal ablation.* The paper reports that ablating cross-attention to up-steps costs about 1.5% exact-match accuracy, read as evidence of a selective mechanism attending preferentially to up-steps. We measure a baseline of 0.996 exact match falling to 0.0000 when ablating attention to up-steps, and to 0.0005 when ablating attention to the complementary down-steps. Both ablations are catastrophic and symmetric: cross-attention is load-bearing wholesale in our model, with no selectivity toward either step type.
- *Attention statistics.* Mean cross-attention mass on up-steps is 0.5344 against a uniform baseline of 0.5; the attention argmax lands on an up-step for 0.5739 of queries; only 0.0682 of queries have their argmax at the path’s own maximum level; the mean level of the argmax-attended position is 2.369, against 2.471 averaged over all source positions. These are essentially uniform statistics – we find no evidence of attention selecting the path’s highest level.
- *Linear probes.* The paper reports approximately 84% probe accuracy for recovering path levels from encoder hidden states. We measure 0.9478 (14 classes, majority baseline 0.2439) for level, and 1.0000 for step type (2 classes, baseline 0.5). This one direction replicates, and in fact strengthens.
- *Length generalization.* The paper evaluates lengths $n = 11$ –16. Our model scores exactly 0.0000 exact match at $n = 11, 12, 14, 15, 16$, with per-token character accuracy 0.5872

at $n = 12$ against roughly 0.5 chance. We tried four positional-placement variants at evaluation time (left-aligned to length 24, left-aligned padded to length 26, right-aligned, centered); all four give 0.0000 exact match, which rules out a pure positional-embedding-range explanation for the failure.

We draw the conclusion carefully, and without asserting that the original authors made an error: same architecture, same task, the same held-out accuracy we would need to call the reimplementations faithful, and yet a materially different internal mechanism by every measure we could apply. Whatever mechanism a small transformer settles on for a mathematical computation appears to be, at least in part, a property of the particular training run rather than a property of the task it was trained on. For an interpretability-driven discovery programme this is the central methodological risk: an extracted “mechanism” may describe a training run rather than a fact about the mathematics. This is exactly the failure mode we then observed inside our own setting, at seed 3 of §8.4.

8.6 Assessment

None of the four contributions of this paper came from the neural models. The one thing the machine-learning line delivered that fed forward into the rest of the paper was the degree-5 arrangement feature family of §8.4, which sharpened our attention on H_5 and became one input, among others, to the rule-mining process that produced the candidate rules discussed in Section 9 – of which the surviving rule, in its exhaustively verified form, became Theorem 4.1. We describe that as an indirect, heuristic contribution and make no stronger claim than that.

We also record two structural failures of the programme, so that the positive results above are read in proportion. First, the dynamics model – the entire interpretability design bet, and the closest analogue to the zeta-map paper’s step-by-step mechanism – attains teacher-forced verdict accuracy 1.0000, but collapses under free-running rollout to 0.7778 at $r = 9$, 0.7786 at $r = 10$, and 0.6245 at $r = 10$ in the second ($r10v2$) configuration, with free-running boundary accuracy exactly 0.0000 at $r = 10$ and $r10v2$: unrolled without teacher forcing, the model converges to the trivial fixed point of predicting that nothing is ever eliminated. Second, the D1 checkpoint’s learned positional embeddings extend only to length 259, which admits 7,021 of the 14,036 configurations in the corpus; ring sizes above roughly 13 were structurally out of the model’s reach, independent of anything about the underlying mathematics.

Finally, we caught and quarantined one tautological feature during this work. The quantity $n_{\text{consistent}}$ correlates with the verdict label at Spearman -0.850 and appears to “close 167% of the gap” to perfect boundary accuracy – because, on inspection, it is the label in disguise: $n_{\text{consistent}} \leq 50$ is one of the two disjuncts defining boundary non-reducibility in the first place. It is excluded from consideration via `TAUTOLOGICAL_CANDIDATES` in `tools/d1_interrogate.py`, and we mention it here only as a caution to anyone repeating this kind of search: a feature that predicts the label unusually well is worth checking for circularity before it is reported as a discovery.

9 Negative results and killed claims

We report these at length because the verification discipline of §7 is only credible if its failures are visible, and because each of the three killed claims below was, at the moment it was killed, the most attractive result the project had.

9.1 Three mined sufficient conditions, killed by mutation

We mined a pool of 16,794 canonically deduplicated configurations (11,919 D-reducible / 4,875 not, rings 6–16, 0 label conflicts) for minimal conjunctions of structural predicates that are sufficient for D-reducibility with zero false positives on the pool. The search space contains 81,386 such minimal zero-false-positive rules. The three best by coverage were:

#	rule	coverage	pool FPs
1	$n \geq 25$ AND $\overline{\text{deg}}_{\text{int}} \geq 5.7\overline{3}$ AND $h_5\text{-density} \leq 2/7$	2,313	0
2	$n \geq 25$ AND max run of constant ring degree ≤ 4 AND $n_{\text{deg } 5 \rightarrow \geq 3 \text{ deg } 5} = 0$	2,240	0
3	$k \geq 11$ AND $h_5\text{-density} \leq 2/7$ AND $n_{\text{deg } 5 \rightarrow \geq 3 \text{ deg } 5} = 0$	2,234	0

All three survived 5-fold cross-validation, but that survival was already weak evidence: across the five folds their coverage ranks are 1, 1, subsumed, 38, 1 for rule 1, 2, 2, subsumed, 47, 7 for rule 2 and 12, 4, subsumed, 43, 4 for rule 3 — rule 3 never reaches first or second in any fold, and all three fall far down in fold 4. All three are **false**.

Adversarial mutation search (§7, M2) generated 339,600 mutants from 9,391 seeds; 248,861 failed the structural legality gate, 88,496 no longer satisfied the rule, 291 were duplicates, and 1,952 novel legal rule-satisfying configurations were submitted to the reducibility checker. Of those, 236 are **false positives** — configurations satisfying the rule that are not D-reducible: 83 for rule 1, 44 for rule 2, 109 for rule 3, all with distinct canonical keys, at rings 13 (54), 14 (105), 15 (56) and 16 (21), arising from 113 distinct seed configurations. The false-positive rate among labelled mutants is 12.1%. 215 of the 236 were **independently confirmed** by the compiled C oracle, with 0 disagreements between the two implementations; the remaining 21 — exactly all the ring-16 cases — could not be adjudicated because the oracle’s output truncates before its verdict line, the same failure mode as §5.

The lesson, stated as we now apply it: zero false positives on a pool of 16,794 configurations, stable across cross-validation folds, is not evidence of soundness. It is evidence that the pool does not contain the counterexamples, which in a curated catalog is what one should expect. Always mutate-stress.

9.2 The rule that survived — and why the mutation test did not really test it

The dual, *necessary*-direction rule — “ $r = 11$ and $k \leq 6$ implies not D-reducible”, covering 1,630 non-reducible configurations with zero false positives — was the one candidate the mutation search did not kill, and it is the statement that became the $r = 11$ case of Theorem 4.1.

Its “survival” should nevertheless be discounted almost entirely. Of 41,449 mutants generated, 41,148 failed the legality gate, 273 failed the rule and 28 were duplicates: **zero novel rule-satisfying mutants were ever submitted to the checker**, and the run took one second. The tool’s verdict is computed mechanically as “killed if any false positive, else survived”, so SURVIVED here means “no counterexample was found”, not “counterexamples were sought and none exist”. The reason is structural and instructive: configurations at $r = 11$ with $k \leq 6$ sit on the legality boundary — their interior vertices have degree exactly 5 — so almost every single edge flip drops a vertex below its minimum degree. Raising the per-seed budget to 400 tries still yielded nothing testable.

The real evidence for that statement is therefore not the mutation test but the exhaustive `plantri` sweep of §4.2. We flag this because presenting the mutation result as independent confirmation would be exactly the kind of thing this section exists to prevent.

9.3 The strong LP encoding, killed by its own negative control

The natural formalisation of “the discharging closes” — *every unblocked stage-1 wheel must end with strictly negative charge* — is wrong, because the 2026 schema permits survivors and the published proof leaves 16,148 stage-1 wheels alive on purpose. We implemented it anyway, with the max-of-affine cutting-plane machinery it requires, and ran the *full-pool* control first. The control is INFEASIBLE at iteration 0 with a verified 10-row certificate ($y^\top b = -175$): the encoding “disproves” the published theorem. We discarded the encoding on that basis and built the leaf-level one of §3 instead.

We stress the ordering. The control was run *before* the $\text{ring} \leq 14$ variant, and the $\text{ring} \leq 14$ variant under that encoding is also infeasible — with a nicely large $y^\top b = -380$. Had we run the two in the other order and stopped, we would have published a false result with a valid-looking certificate. A negative control that can fail is the only defence against this, and it is cheap.

9.4 The naive threshold law, killed by the harness

The first form of the threshold half of Conjecture 6.1, read off the $r = 8$ minimum and the observed growth ratio, was K D-reducible $\Rightarrow a \geq 94 \cdot 2.4^{r-8}$. The harness kills it with 43 counterexamples in about 0.2 s. Also killed: $a \geq 2.4^r/12$ (15 counterexamples, and still 2 counterexamples when restricted to the machine-generated population, so this is not a catalog artifact); $a \geq 5 \cdot 2^{r-4}$ (6); $a \geq 2^{r-1}$ (311); $a \geq k \cdot 2^{r-4}$ (398); $a \geq W(r) \cdot 1.3^{k-1}$ with no leading constant (all 13,169); $a \geq 3^{r-1}/2/(0.25r^2)$ (17). On the cap side the harness killed $a \leq 5 \cdot 2^{r+k-6}$ with exactly six counterexamples — the six wheels — which is precisely what Theorem 6.4’s min-over- h form predicts, and is the reason the unconditional constant is 11 and not 10. The surviving forms in §6 are what is left after that attrition.

9.5 Two further honest notes

A conjecture we would have overclaimed. An earlier draft of §6 stated $\gamma = 3/2$ for all k , under a hypothesis (H2) about arc lengths that looked innocuous. (H2) fails on at least 53,549 of 59,142 records ($\geq 90.5\%$) and on 37.66% of individual BFS steps (140,084 of 372,008). The adversarial reviewer called presenting the conclusion under that hypothesis “a real overclaim”, and it was demoted to Theorem 6.5.

Corrections to our own record. Two numbers circulated in this project’s internal notes that are wrong and that we correct here: the mutation search produced 236 false positives, not 26 (an apparent dropped digit, propagated through three documents); and the wheel-candidate total at hub degrees 7–11 is 5,692,937, not 5,690,937. Both are corrected in this paper and neither affects a verdict.

10 Discussion and open problems

The sharp cap via transfer matrices. Theorem 6.6 says the sharp cap cannot be reached through $P(S, 4)$; one needs a direct handle on $|\Phi(K)|$. Two routes remain. (a) An injection $\Phi(K) \hookrightarrow \Phi(K') \times [4/3]$ for a configuration K' with one fewer interior vertex *and the same ring*. Ordinary vertex deletion will not do: deleting an interior vertex of degree d with j consecutive ring neighbours yields ring length $r + d - 2j + 2$, so the ring *grows* — for $d = 5$, $j = 2$ it grows by 3, multiplying the cap by 8, and the induction would have to supply $a(K) \leq a(K')/6$, an inequality of the wrong strength. Deletion-based induction on this class is a dead end unless the ring can be kept fixed. (b) A transfer-matrix bound on the *ring-trace* operator, whose state space is ring colourings rather than tri-colourings. The Jacobsen–Salas–Sokal Temperley–Lieb machinery [12] is the right formalism, and its state space — non-crossing partitions of the ring — is exactly the object Φ lives on. We regard this as the highest-value open problem in the programme. Failing that, proving $\gamma = 3/2$ in general (H2 is a finite, checkable link condition) already lifts the implied f -slope from 0.263 to 0.450 and is permitted by the Bridge route.

Removing the fixed-shapes caveat. Theorem 3.1 fixes the 84 published rule shapes. The natural strengthening — “no discharging schema of bounded rule complexity closes over the $\text{ring} \leq 14$ pool” — requires column generation: given a violated leaf, price out a new planar degree-interval pattern that would fix it. The pricing subproblem is a search over planar patterns with a charge dart, not an LP, and it is the main technical obstacle. A tractable intermediate is to enumerate all rule shapes below a fixed size bound and re-run the same certificate machinery over the enlarged column set: infeasibility there would be a substantially stronger statement, and the certificate format is unchanged.

$f(r)$ beyond $r = 13$. The $r = 13$ interior-7 cell alone required 11,534,001 plantri graphs and ≈ 30 core-hours, 8-way sharded (≈ 3.7 h wall); $r = 14$ at interior 8 is roughly two orders of magnitude larger and is not reachable by the present pipeline on a workstation. Three ways forward: (i) prune the generation with a necessary condition on a — the mass law, if proven even in weak form, would eliminate most cells without running the reducibility check; (ii) exploit the observation of §6 that the minimum- a and minimum-interior frontiers coincide, to search only near the frontier; (iii) accept a conditional result. Note that the corpus already gives $f(14) \leq 9$, $f(15) \leq 10$, $f(16) \leq 10$ from catalog witnesses.

Formalisation. The finite content of Theorem 4.1 is a replay of 10,319 configuration checks. That is within reach of a proof assistant given Gonthier’s formalisation [9] of the reducibility machinery, and it would upgrade the theorem from “certified” to “proven” in the strongest available sense. The Farkas certificate of Theorem 3.1 is even more tractable: 12 rows over the rationals, though the row-derivation step (from cartwheel to coefficients) carries the real formalisation burden.

The set-size question. Our original motivation was to minimise the number of configurations in an unavoidable set. We deliberately do not report a headline number, because a minimal U in the wheel-based schema is *not* comparable to RSST’s 591/633 (axle-based, C-reducibility with contracts, $\text{ring} \leq 14$). Any such claim must be schema-qualified. Theorem 3.1 is the useful residue of that line: an impossibility rather than an improvement.

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A Claim-to-artifact table

Every numerical claim in the paper is listed here with the repository artifact that produced it. Paths are relative to the repository root. A claim marked CORPUS is a lemma-harness verdict over the 59,142-configuration labelled corpus with signature `a2b1feba331bf2962b2b` and carries a receipt in `results/mass-law/lemma.log.jsonl`; CORPUS means “survived”, never “proved”.

Claim	Artifact and content
§3: the $\text{ring} \leq 14$ obstruction 84 rule shapes; $r(R) \in \{1, 2\}$	<code>third_party/discharging-rules/R/</code> (84 <code>.rule</code> files); <code>third_party/arxiv-2603.24880/src/main_1_1.tex</code> lines 624–625
8,200-config pool; 5,895 of $\text{ring} \leq 14$	<code>third_party/reducible-configurations/D/</code> (8,200 files); <code>build/steinberger-pool-r14/D</code> (5,895); ring histogram recomputed from the N R header of every file
Affine leaf form validated on 10,094 published leaves, 0 discrepancies 8-variant LP matrix (Table 1)	<code>tools/lp_discharge.py --validate</code> ; leaves in <code>third_party/computer-checks/wheels/zero/</code> <code>results/steinberger/lp_variants_summary.json</code> , <code>results/steinberger/lp_control_stats.json</code> , <code>results/steinberger/lp_main_stats.json</code>
19-row Farkas certificate, $y^\top b = -30$ 12-row irreducible core, $y^\top b = -20$	<code>results/steinberger/lp_main_certificate.json</code> <code>results/steinberger/lp_main_certificate_iis.json</code>
Exact-rational verification, rows re-derived	<code>tools/verify_farkas.py</code> $\rightarrow$ FARKAS CERTIFICATE VERIFIED, exit 0
All 12 rows mathematically necessary	<code>results/steinberger/iis_math_necessity.json</code> , <code>all_mathematically_necessary: true</code>

Claim	Artifact and content
11 of 5,292 (rule, out-dart) pairs undetermined	tools/lp_discharge.py --audit-necessity
Low-degree refinement counts (4 necklaces, up to 2.4×10^8) and the 3,046-wheel sweep	tools/lowdeg_refinement_check.py, results/steinberger/lowdeg_refinement_check.json
580 / 2,466 unblocked $d=5$ / $d=6$ wheels, matching the C++	released C++ --enum_wheels -d 5, -d 6; same counts in the low-degree sweep above
Textual audit of the $d \leq 6$ reconstruction	results/steinberger/degree56-verification.md (quote-by-quote against the pinned paper source)
Strong-encoding negative control: full pool INFEASIBLE, $y^\top b = -175$	tools/lp_wheel_level.py --control, results/steinberger/wheel_level_control_stats.json
77,185 leaves recovered from 833 aborted jobs	tools/lp_rerun_failed_wheels.sh, results/p3/runs/steinberger-s1/work/wheels/zero_ndebug/, results/steinberger/rerun_ndebug.log
S1: 833 of 16,157 jobs abort; 780 C.positive + 53 degree_range; 530/250/51/2 by hub degree	results/p3/runs/steinberger-s1/failures.json; job counts from the .done/.FAILED markers under results/p3/runs/steinberger-s1/log/
Full narrative, scope and adversarial pass <i>Narrative-only</i> (recorded in prose in LP-SCHEMA-RESULT.md, no JSON): the 2,145 monotonicity parent/child pairs and their 8 failures under negative amounts; the six solver configurations and the 0.082 uniform slack; the degree-5-alone / degree-6-alone ablations; the ≈ 14 min / ≈ 2 GB cost of --all	results/steinberger/LP-SCHEMA-RESULT.md re-derivable only by re-running tools/lp_discharge.py; we flag these as the claims in §3 not backed by a stored machine-readable artifact
§4: the $f(r)$ theorem	
Per- (r, n) sweep counts (Table 2)	results/theorem/fr_table/manifest_r{8..13}.json (r13 additionally _s{0..7}of8); every entry has d_reducible_count: 0
Per-configuration records (adjacency, a , closure trace, legality flag)	results/theorem/fr_table/configs_r*n*.jsonl
Cross-check vs. the compiled 1995 oracle: 70 sampled, 0 mismatches, 6 inconclusive	results/theorem/fr_table/crosscheck.json; tools/fr_table_crosscheck.py
Third-implementation adjudication of the 6	src/fourcolor/brute_force.py, tests/test_brute_force.py
Tightness witnesses at $r = 8..12$	results/theorem/fr_table/tightness_witnesses.json
Tightness at $r = 13$ ($k = 8$, $a = 6,954$) and the per-ring minimum- k table	the labelled corpus via src/fourcolor/lemma_corpus.py
C-vs-D witnesses at rings 8 and 11	third_party/arxiv-1401.6481/src/anc/unavoidable.conf idents 2.126, 126.7566, 128.136, 7562.126
Mass-law gap table (§4.5)	recomputed from results/theorem/fr_table/configs_r*.jsonl (max a over legal below-threshold configs) against the corpus minima
Enumeration completeness	third_party/plantri/plantri55/ with sha256 of binary and source; legality by four-color.mutate.is_legal_configuration

Claim	Artifact and content
Statement and protocol	results/theorem/fr_table/THEOREM.md
<i>§5: corpus verification</i>	
RSST 633: 633/633 agree; 249 D-reducible, 384 C-reducible	results/differential-unavoidable-r14/report.jsonl; results/oracle-rsst-633/stdout.txt
Steinberger 2,822: 2,822/2,822 agree, all D-reducible, all $b = 0$	results/differential-U.2822-r16/report.jsonl, results/differential-stein-log.txt
The Steinberger C oracle aborts at config 783	results/oracle-stein-2822/stdout.txt (ends exit=134); crashing input build/crash783.conf
7,697 of the pruned pool verified, 73 at rings 17–18 unverified	results/p3/pool-verify/SUMMARY.json, shard_r*.jsonl
2026 pipeline reproduced end to end, all counts matching	results/nl4ct-reproduction-summary.txt;
Differential on all 5,692,937 wheel candidates, 0 FP / 0 FN	third_party/computer-checks/log/ results/nl4ct-differential/report_d{7..11}.json (exact_match: true in each)
RSST 1995 discharging replay, 5/5 presentation files	results/discharge-rsst/replay.txt
Wheel-level coverage: 85 distinct blockers; cartwheel-level: 7,422 over 2,253,185 events	results/p2-coverage/report.md, results/p2-coverage/wheel_level_stats.json, results/p2-coverage/cartwheel_level_stats.json
430 zero-usage configurations	results/p3/usage_table_v2.summary.json, results/p3/zero4_configs.txt (430 entries)
Pruned pool of 7,770 passes the whole pipeline with identical counts	results/p3/runs/batch1b/ — VERDICT.txt (PASS), counts.json, checksums.sha256
The failed 673-config prune that revealed the fourth role	results/p3/runs/batch1/VERDICT.txt (FAIL, rc 134)
<i>§6: the coloring-mass law</i>	
Corpus of 59,142, signature a2b1feba331bf2962b2b, 13,169 D-reducible	src/fourcolor/lemma_corpus.py (recomputed live); results/mass-law/lemma.log.jsonl
Wheel closed form ($k = 1$, support 6)	CORPUS receipt b2fb2220ca3e6953; proof in §6; four-way adversarial re-verification in results/mass-law/PROOF-SHARP-CAP.md §6.3
$a \leq 11 \cdot 2^{r+k-7}$; min-over- h form tight only at the six wheels	receipts e8beefcd5e56f5c5, 4db503ddffcef7ee; 9977aea202daca71 (the killed $5 \cdot 2^{r+k-6}$)
Star-order claim verified on all 431,159 (record, interior vertex) pairs	tools/star_order_check.py, results/mass-law/star-order-verification.md
Exactly two common neighbours on all 610,846 interior–interior edges	receipt 663fe16bb1a8ea3c
(H2) false: arc histogram, $\max \gamma_{\text{prov}} = 1.937082$	tools/arc_profile.py, results/mass-law/arc-profile.md
Obstruction theorem: $P(S_0, 4) = 1464$, $a = 55$ at gen-r8-n10-2; P -form killed on 1,960 of 59,142	src/fourcolor/count4.py, tools/p4_measure.py, results/mass-law/p4-summary.md, results/mass-law/p4_measurements.jsonl
Per-step factor table ($b_i = 3$ mean 1.3155, max 1.6479)	results/mass-law/star-order-verification.md Part 4
Lattice disks: P -form fails by 1.1×10^4 ; a satisfies the sharp cap with improving margin	tools/lattice_disk_entropy.py, tools/lattice_disk_phi.py, results/mass-law/lattice-disk-{entropy,phi}.md

Claim	Artifact and content
Threshold frontier, $\beta = 2.4$ residual band, (T1) optimal to 0.08%	results/mass-law/threshold-candidates.md (75 receipts)
Implied $f(r)$ tables	results/mass-law/threshold-candidates.md §4, results/mass-law/PROOF-SHARP-CAP.md §7
Weak cap and Bridge Lemma	results/theorem/mass-law/PROOF-CAP.md
<i>§8 and §9</i>	
D1 training and probe ladder	results/d1/train_metrics_full_v1.json, results/d1/interp/report.md re-
Boundary result, extraction, seed replication, error-set digestion	results/d1v2/v2run/interrogation.md, results/d1v2/extraction/report.md, results/d1v2/replication/report.md, results/d1v2/digestion/cases.md re-
Zeta-map replication (companion repository)	companion repository zeta_map.interp: results/train-metrics.json, results/generalization.json, results/interp/{probes,ablation,attention-stats}.json re-
Mined candidate rules, coverage, cross-validation	results/theorem/candidates.md, results/theorem/mining_results.json; tools/mine_sound_rules.py re-
Mutation stress test: 339,600 mutants, 1,952 labelled, 236 FPs, 215 oracle-confirmed	results/theorem/stress_test.json, results/theorem/stress_test.md (every FP carries full adjacency and feature vector); tools/stress_test_theorem.py re-
Killed threshold forms	CORPUS receipts in results/mass-law/lemma_log.jsonl; tabulated in results/mass-law/threshold-candidates.md

B The Farkas certificate

Table 8 gives the 19-row support of the certificate for Theorem 3.1. Exact totals: $\min_k(y^\top A)_k = 0$ (need ≥ 0), $y^\top b = -30$ (need < 0), all $y_i > 0$. Every row is non-strict, so the certificate proves infeasibility over $\mathbb{R}_{\geq 0}^{84}$ and not merely over the integers. The greedy-reduced 12-row irreducible core is marked in the IIS column, with its own multipliers y' ; it has $y'^\top b = -20$ and is irreducible in the strict sense that deleting any single one of its 12 rows restores feasibility (verified for all 12). Machine-readable: results/steinberger/lp_main_certificate.json and results/steinberger/lp_main_certificate_iis.json; verify with tools/verify_farkas.py.

Table 8: The 19-row Farkas support ($y, y^\top b = -30$) and the 12-row irreducible core ($y', y'^\top b = -20$; a dash marks a row absent from the core). $C(x_0)$ is the leaf charge at the published amounts; a positive value marks a row the published amounts violate. All four ring- ≤ 14 -only rows survive into the core.

y_i	y'_i	d	spoke degrees	rhs	$C(x_0)$	source	origin
19	75	7	5,5,8,5,6,5,7	10	0	d7_1155_1	full $\cap$ r14
21	—	7	5,5,5,8,5,5,7	10	0	d7_166_0	full $\cap$ r14
17	75	7	5,5,5,8,5,5,7	10	0	d7_166_1	full $\cap$ r14
14	—	7	5,6,5,6,7,5,8	10	0	d7_1782_2	full $\cap$ r14
12	—	7	5,6,5,8,5,7,6	10	0	d7_1932_3	full $\cap$ r14
4	25	7	5,5,7,5,5,7,7	10	0	d7_710_10	full $\cap$ r14
4	—	7	5,5,7,5,5,7,7	10	0	d7_710_14	full $\cap$ r14
21	75	7	5,5,7,5,6,5,8	10	0	d7_722_3	full $\cap$ r14
9	—	7	5,5,7,5,7,5,7	10	0	d7_740_16	full $\cap$ r14
8	—	7	5,5,7,5,7,5,7	10	0	d7_740_18	full $\cap$ r14
13	55	7	5,5,7,5,7,5,7	10	0	d7_740_19	full $\cap$ r14
20	60	7	5,7,5,7,6,7,7	10	2	d7_3358_2133	ring≤ 14 only
6	15	7	5,7,5,7,6,7,7	10	2	d7_3358_2134	ring≤ 14 only
30	55	7	5,7,5,7,7,6,7	10	2	d7_3372_1881	ring≤ 14 only
4	25	7	5,7,5,7,7,7,7	10	3	d7_3376_1132	ring≤ 14 only
2	—	5	5,5,6,7,7	-10	-3	d5-55677	low-degree row
43	75	5	5,5,7,6,7	-10	-2	d5-55767	low-degree row
75	168	5	5,7,5,7,7	-10	0	d5-57577	low-degree row
85	219	5	5,7,6,7,7	-10	0	d5-57677	low-degree row